

Degree Project in Electrical Engineering

Second cycle, 30 credits

Design Research Case Study of LLM-Based Conversational Sustainability Reflection System for Creative AI Practitioners

SHAOTIAN WU

Design Research Case Study of LLM-Based Conversational Sustainability Reflection System for Creative AI Practitioners

SHAOTIAN WU

Master's Programme, Information and Network Engineering, 120 credits
Date: September 5, 2025

Supervisor: Petra Jääskeläinen
Examiner: Pawel Andrzej Herman
School of Electrical Engineering and Computer Science
Swedish title: Fallstudie inom designforskning av ett LLM-baserat konversationellt hållbarhetsreflektionssystem för kreativa AI-utövare

© 2025 Shaotian Wu

Abstract

As the use of generative AI tools grows across creative domains, so too does the need to reflect on their environmental impact. This thesis presents a research through design study that involves design and evaluation of a conversational sustainability reflection system, aimed at supporting creative practitioners who use AI considering sustainability trade-offs within their workflows. Grounded in principles from behavior change, explainable AI, and interactive reflection, a chatbot prototype that engages users in emotionally supportive and non-judgmental dialogue was developed. The design process followed an iterative, research-through-design approach, incorporating thematic analysis from surveys and user testing with creative AI practitioners. Based on survey insights, the design focused on embedding low-pressure, emotionally supportive prompts and summary-style sustainability feedback into creative workflows. Key findings from the qualitative user study further revealed that users appreciate empathetic prompting and low-pressure framing, which foster reflection without inducing moral fatigue. However, challenges remain in the timing, clarity, and personalization of sustainability feedback, as well as in maintaining conversational continuity. The study also found that users are more likely to adopt the tool if it is embedded into existing creative platforms, rather than used as a standalone application. The results suggest that conversational agents can serve as effective mediators of value-sensitive reflection, provided they are context-aware, emotionally attuned, and technically unobtrusive. Future research may explore personalization techniques, long-term engagement, and deeper integration with creative ecosystems.

Keywords

Sustainability Reflection, Conversational AI, Creative AI, Reflective Design

Sammanfattning

I takt med att generativa AI-verktyg blir allt vanligare inom kreativa områden, ökar också behovet av att reflektera över deras miljöpåverkan. Denna uppsats presenterar en designforskning (research through design) som omfattar utformning och utvärdering av ett konversationsbaserat reflektionsverktyg för hållbarhet, med syftet att stödja kreativa yrkesutövare som använder AI i sitt arbete och vill väga in hållbarhetsaspekter. Baserat på principer från beteendeförändring, förklarbar AI och interaktiv reflektion, utvecklades en chatbot-prototyp som engagerar användaren i ett emotionellt stödjande och icke-dömande samtal. Designprocessen följde en iterativ forskningsbaserad metod, med tematiska analyser av enkätdata och användartester med kreativa AI-praktiker. Utifrån enkätresultaten fokuserade designen på att integrera lågtröskliga, emotionellt stödjande uppmaningar samt sammanfattande återkoppling om hållbarhet i kreativa arbetsflöden. Den kvalitativa användarstudien visade att deltagarna uppskattade empatiska uppmaningar och lågtrycksramar, vilket främjar reflektion utan att leda till moralisk utmattning. Samtidigt framkom utmaningar gällande återkopplingens timing, tydlighet och grad av personalisering, samt behovet av bättre kontinuitet i samtalen. Studien visade också att användare är mer benägna att ta till sig verktyget om det integreras i redan befintliga kreativa plattformar, snarare än att användas fristående. Resultaten tyder på att konversationsagenter kan fungera som effektiva medier för värdebaserad reflektion, under förutsättning att de är kontextmedvetna, emotionellt lyhörda och tekniskt diskreta. Framtida forskning kan undersöka personaliseringstekniker, långsiktig användarengagemang samt djupare integration i kreativa ekosystem.

Nyckelord

Hållbarhetsreflektion, konversations-AI, kreativ AI, reflekterande design

Acknowledgments

I would like to express my heartfelt gratitude to my supervisor, Petra Jääskeläinen, for her detailed suggestions and patient communication. Her invaluable guidance and continuous support have been instrumental throughout the entire thesis process. I am also grateful to my examiner, Pawel Herman, for his thoughtful feedback and kind assistance during my thesis work. My deepest thanks go to my parents. Their unwavering encouragement and support have made this journey possible. I also wish to thank all participants who took the time to complete the survey and join the user study. Their generous feedback and suggestions laid the foundation for this thesis. Finally, I would like to thank myself—for staying committed, curious, and resilient in completing this meaningful and engaging thesis project at the end of my Master's studies.

Stockholm, September 2025
Shaotian Wu

Contents

1 Introduction **1**

2 Background **4**
2.1 Sustainability in Creative AI 4
2.2 Explainable and Conversational AI for Sustainability 7
2.3 Behavior Change and Reflection in AI-Driven Creativity . . . 8

3 Methodology **11**
3.1 Research Approach 11
3.2 User Survey and Needs Assessment 11
3.3 Prototype Development 13
3.4 Experimental design/Planned Measurements 14
3.5 Ethical Considerations 18
3.6 Limitations 18

4 Sustainability Reflection Chatbot System Design **20**
4.1 Post-Generation Reflection through Ambient Micro-Prompts . 21
4.2 Emotionally Supportive and Empathetic Prompting Tone . . . 22
4.3 Summary-Based, Situationally Embedded Sustainability Feedback 25

5 Results **29**
5.1 Survey Insights 29
5.1.1 Survey Questions Analysis 29
5.1.2 Survey Summary 43
5.2 User Study Results 43
5.2.1 Quantitative Results 43
5.2.2 Qualitative Results 48
5.2.2.1 Ambiguous Onboarding: From Confusion to Recognition 48

5.2.2.2 Reflection Resistance: When Users Feel They've Already Done Enough 49
5.2.2.3 Broken Flow: Memory Gaps and Perceived Disconnection 51
5.2.2.4 Information Depth vs. Emotional Softness: Struggling to Satisfy Both 52
5.2.2.5 Design Suggestions: Personas, Integration, and Closure 53
5.2.2.6 Cultural and Demographic Observations . . 54

6 Discussion **57**
6.1 Summary of Findings 57
6.2 Reflection on the Design Process 59
6.2.1 Reflections on the theoretical framework 59
6.2.2 Reflections on the design specification 60
6.3 Future Work 61

7 Conclusions **63**

References **65**

A User test plan **76**

B Consent form of the User Study **80**

C AI Tools Usage Statement **82**

List of Figures

4.1 System state immediately after content generation 23
4.2 Post-generation micro-prompt asking for response evaluation . 23
4.3 Simplified interaction flow for post-generation feedback 23
4.4 Example of empathetic reflection prompt phrased through prompt-level tone conditioning 25
4.5 Example of context-aware, embedded sustainability prompt . 27

5.1 The answer of Q1 44
5.2 The answer of Q2 44
5.3 The answer of Q3 45
5.4 The answer of Q4 45
5.5 The answer of Q5 46
5.6 The answer of Q6 46
5.7 Individual scores from participants 47

List of Tables

3.1 The participant demograpic information 15

4.1 Different questions in case of different tones 24

Chapter 1

Introduction

The exponential growth of artificial intelligence (AI) in creative fields has revolutionized artistic practices, enabling novel forms of expression through generative models such as DALL-E and Stable Diffusion [1, 2]. However, this rapid advancement comes at a significant environmental cost. Studies have shown that training and deploying large-scale AI models require vast computational resources, leading to substantial energy consumption and carbon emissions [3, 4]. While sustainability has become an increasing concern across AI research, creative AI practitioners—referring to artists, designers, and developers who use AI models such as image generators or chatbots in their creative process—often lack accessible tools to assess and reflect on the environmental impact of their work [5]. This gap raises critical questions about how to integrate sustainability awareness into AI-driven creative workflows.

Existing research in Human-Computer Interaction (HCI) has explored sustainability reflection systems, particularly in areas such as energy monitoring and eco-feedback systems [6, 7]. However, little work has been done to extend these efforts to creative AI systems. Notably, P. Jääskeläinen et al. [5] highlight the urgency of embedding sustainability considerations within creative AI practices, yet practical implementations remain scarce. Emerging projects like The Green Notebook [8] and other sustainability-focused AI interventions [9] demonstrate the potential of conversational AI interfaces in engaging users with sustainability reflection. These studies suggest that dialogue-based interactions can encourage deeper awareness and consideration of sustainability trade-offs within creative processes.

Building upon this foundation, this study investigates how large language models (LLMs) can serve as interactive agents for sustainability reflection

in creative AI. In this thesis, sustainability reflection refers to the process of encouraging users to become more aware of the environmental implications of their creative choices and to consider alternative actions within their workflows. The potential of LLMs lies in their ability to facilitate dynamic, personalized, and context-aware discussions that prompt users to critically evaluate their choices. Research in behavior change suggests that passive presentation of sustainability data is insufficient to drive meaningful action; instead, interactive and engaging approaches are needed to foster reflection and decision-making [10, 11]. Conversational AI presents an opportunity to bridge this gap by offering an intuitive and adaptive medium for sustainability engagement within AI-driven creative workflows. In this thesis, sustainability reflection refers specifically to reflective thinking that enables practitioners to make more informed and sustainable creative decisions—such as reusing generated assets, adjusting prompts, or reconsidering high-cost outputs—rather than policy or infrastructural interventions in the broader creative AI ecosystem.

This study follows a Research-through-Design (RtD) approach [12] to develop and evaluate a conversational sustainability reflection system for creative AI practitioners. A user-centered design process (e.g., [13]) was employed, involving user research, iterative prototyping, and user testing to ground the system in real-world needs. The system design involves a set of dialogue-based interactions designed to support sustainability reflection. These interactions were guided by three key design features obtained through a user research survey: ambient micro-prompts after generation, which encourage reflection without disrupting the creative flow; empathetic and emotionally supportive tone, to maintain a non-moralistic and non-judgmental atmosphere; and summary-based feedback, which presents sustainability insights in a concise and contextualized manner rather than through raw data or metrics.

The user test findings suggest that such a system can increase sustainability awareness and encourage reflective thinking in creative workflows.Participants appreciated the tool's supportive tone, subtle prompts, and contextualized summaries, which collectively helped them to momentarily pause and re-evaluate their practices. While direct behavioral change was not always observed, the tool fostered a space where users could revisit their assumptions and consider trade-offs more consciously. While this thesis focuses on the initial design and evaluation of a reflection-oriented chatbot, future research may build on this foundation by exploring more advanced personalization techniques, such as adaptive prompting based on user behavior and

preferences. This direction is motivated by participant feedback from the user test suggesting that more tailored and context-aware responses could further enhance engagement and relevance. Additionally, studies on long-term engagement and retention could help determine how conversational sustainability support might evolve over time, especially given that several participants expressed interest in using such a tool on an ongoing basis. This suggests that understanding how users engage with reflective prompts over extended periods could inform more sustained and meaningful support strategies. Finally, closer integration into existing creative AI ecosystems—such as image generation platforms or writing assistants—could offer more seamless and practical adoption paths. This insight emerged from participants' comments suggesting that embedding the chatbot within their existing creative workflows would reduce friction and enhance usability.

The structure of this thesis is as follows: Chapter 2 provides the theoretical background on sustainability in creative AI, explainable and conversational AI for sustainability and behavior change and reflection. Chapter 3 describes the methodology, including data processing, model implementation, and evaluation. Chapter 4 presents the results from the experiments. Chapter 5 discusses the findings and limitations of the study. Chapter 6 discusses about the reflection on the design process and the future work remaining to be done. Finally, Chapter 7 concludes the thesis with a summary of the contributions and directions for future work.

Chapter 2

Background

2.1 Sustainability in Creative AI

Sustainability is broadly defined as the ability to meet present needs without compromising the ability of future generations to meet their own needs [14]. In this thesis, sustainability is approached through the development of a conversational tool that facilitates sustainability reflection for *creative AI* practitioners, leveraging LLM-based techniques to promote environmentally and socially responsible decision-making. The reflection process in this tool focuses on key areas of sustainability relevant to creative AI, including environmental impact, ethical AI use, and social responsibility.

In this thesis, environmental impact refers specifically to encouraging users to consider the energy consumption, carbon footprint, and resource efficiency related to their AI-generated content — including model training, inference, and data storage [15]. Ethical AI use, as understood in this project, addresses concerns such as bias in generative AI, responsible data sourcing, and the transparency of AI-generated works [16]. Social responsibility, within the context of this thesis, involves reflecting on the broader societal implications of AI-generated content, including questions of inclusivity, accessibility, and potential disruptions to creative industries and labor markets [17]. While this project aims to enhance awareness of sustainability in the creative sector, it acknowledges that certain dimensions, such as economic sustainability or large-scale policy implementation, are beyond its scope. Rather than directly prescribing specific behavioral changes, this thesis emphasizes empowering individuals through reflective tools that prompt critical thinking and self-directed insights. While the ultimate aim of such tools may include encouraging more sustainable behaviors, this thesis focuses

primarily on the design and immediate reflective impact of the intervention, without evaluating long-term behavioral outcomes. By encouraging users to consider the environmental, ethical, and societal dimensions of their creative practices, such tools aim to support more conscious and intentional decision-making within creative AI workflows.

The increasing integration of AI in creative domains raises concerns about sustainability, particularly regarding the computational resources required for generative models. Large-scale AI models consume vast amounts of energy, contributing to their carbon footprint [3, 4]. Despite these concerns, AI-generated art continues to gain popularity, with limited tools available for practitioners to assess their environmental impact. [5] emphasize the importance of incorporating sustainability assessments into creative AI practices. However, few structured frameworks exist to support creative AI practitioners in making informed decisions about energy efficiency and resource utilization. While structured frameworks for assessing sustainability in AI-generated art are limited, a few notable approaches have emerged. Strubel et al. [15] propose energy and carbon footprint estimation methods for AI models, advocating for greater transparency in reporting the environmental impact of training and deploying large-scale neural networks. R. Schwartz et al. [18] introduce the concept of Green AI, which encourages efficiency-oriented research and the development of models that optimize performance while minimizing computational costs. Additionally, [19] presents CarbonTracker, a tool designed to estimate and track the carbon emissions of machine learning models, providing insights into energy consumption and potential mitigation strategies.

Despite these efforts, the integration of such frameworks into creative AI workflows remains limited. P. Jääskeläinen et al. [5] emphasize the importance of incorporating sustainability assessments into AI art practices, yet there is still a gap in accessible and user-friendly tools tailored for artists. The need for structured frameworks that facilitate informed decision-making regarding energy efficiency and resource utilization in AI-meditated creative practices remains an ongoing challenge. A potential challenge in addressing sustainability in creative practices lies in the lack of standardized metrics and accessible frameworks. Many artists and creative practitioners may lack awareness of the energy-intensive nature of AI tools [9], and broader research on AI efficiency has revealed substantial environmental costs even among expert communities [15]. While computational efficiency improvements are being explored at the model level [18], there remains a need for user-facing tools that integrate sustainability reflection directly into the creative workflow.

Recent studies advocate for embedding sustainability prompts within creative software interfaces, encouraging artists to consider environmental impact as part of their decision-making process [5]. For instance, J. Froehlich et al. [6] designed eco-feedback systems that provide users with real-time energy usage data, promoting sustainable behavior. Similarly, T. G. Holmes [7] explored energy-awareness tools in creative environments, demonstrating how users can be nudged towards more energy-efficient choices. Additionally, projects like The Green Notebook [8] have started to integrate conversational AI to prompt users about sustainability considerations during creative processes, showing the potential of such approaches in fostering greater environmental responsibility.

Moreover, sustainability in AI art is not limited to energy consumption alone. The broader lifecycle of AI-generated content—including storage, dissemination, and long-term archival—also poses sustainability challenges [20]. Cloud-based AI services require significant server infrastructure, contributing to carbon emissions through data centers [21]. Studies have shown that data storage and transmission account for a growing share of digital carbon footprints, emphasizing the need for sustainable data management practices [22]. Addressing sustainability in AI art thus requires a holistic approach that considers both the direct and indirect environmental impacts of these technologies, integrating strategies for energy efficiency, responsible data storage, and long-term ecological considerations [19].

Beyond the creative AI domain, Sustainable HCI (Human-Computer Interaction) research provides foundational perspectives for considering technology's environmental and social impacts. Early work in sustainable interaction design emphasized the need to make sustainability a core value in design decisions. For example, Blevis [23] introduced *Sustainable Interaction Design*, advocating for design principles that prioritize resource conservation and reuse in technology development. Subsequent reviews of Sustainable HCI mapped a landscape of approaches ranging from persuasive feedback systems to broader critical and systemic interventions. C.DiSalvo et al. [24] notably *mapped the sustainable HCI landscape*, highlighting how early efforts often focused on individual behavior change through feedback and persuasion, while calling for expanded methods that also consider social, cultural, and systemic factors. This background in Sustainable HCI underlines the importance of embedding sustainability considerations into all stages of technology design and aligns with the goals of this thesis to foster reflection and conscious decision-making in creative AI.

2.2 Explainable and Conversational AI for Sustainability

Human-Computer Interaction (HCI) research has increasingly explored self-reflection systems for sustainability, particularly in areas such as eco-feedback and behavioral nudges [6, 7]. In creative AI contexts, explainable AI (XAI) and conversational interfaces offer promising avenues to support sustainability awareness and engagement. Notably, . Jääskeläinen et al. [5] introduced the concept of explainable sustainability for AI arts, arguing for the importance of transparency in environmental impact and highlighting how creative workflows can benefit from reflective and interpretable sustainability feedback. Studies such as The Green Notebook [8] and interactive sustainability reflection systems [9] illustrate the potential of conversational AI in prompting users to consider environmental implications. These tools leverage interactive dialogues to guide users through sustainability trade-offs, with the aim or potential of fostering a more comprehensive understanding of their impact.

Conversational AI has been explored as a means of enhancing explainability and facilitating user engagement with complex topics such as sustainability [25]. By integrating LLM-powered chatbots into creative workflows, it is of interest to explore how users can receive contextualized sustainability insights without disrupting their artistic process [8]. Unlike static sustainability reports, conversational AI allows for real-time interaction, where users can ask questions, receive tailored recommendations, and reflect on their choices dynamically. Prior research in explainable AI (XAI) highlights that interactive dialogue systems improve user understanding by adapting explanations to their context and needs [26]. This interactive approach aligns with prior work in HCI that suggests reflection is more effective when prompted through personalized and adaptive interventions [10, 11].

Additionally, conversational AI might be able to foster peer-to-peer learning by facilitating knowledge sharing among artists and researchers. AI-powered discussion platforms might have a chance to aggregate insights from sustainability-conscious practitioners, creating a collective knowledge base that supports best practices in sustainable AI artistry. By incorporating social and community-driven reflection mechanisms, conversational AI can further enhance engagement and long-term behavioral change [9].

In parallel, the HCI community has been developing a human-centered approach to Explainable AI that can inform sustainability tools. Researchers

emphasize that explanations should be *intelligible, context-aware, and tailored to users' needs*, rather than merely transparent on a technical level. A. Abdul [27] provide an HCI research agenda for explainable and accountable systems, calling for user-centric explanation interfaces that empower people to understand and appropriately trust AI outputs . From the social science perspective, Miller (2019) synthesizes insights indicating that effective explanations are causal, contrastive, and cognizant of human expectations [28]. Applying these principles in a sustainability context means an AI assistant should explain environmental impacts in relatable terms (e.g., "this action uses as much energy as X") and answer users' follow-up *"why"* questions in a meaningful way. Recent HCI work also expands the notion of explainability to include **social transparency**—making the AI's data sources, uncertainties, or biases visible to users [29]. Such socially grounded explanations can increase users' trust and willingness to engage, which is crucial when prompting artists to reflect on potentially inconvenient sustainability considerations. By leveraging human-centered XAI principles, a conversational sustainability assistant can deliver explanations and prompts that are not only accurate, but also understandable and persuasive for creative practitioners.

2.3 Behavior Change and Reflection in AI-Driven Creativity

Behavior change theories suggest that simply presenting sustainability data is insufficient to influence user decisions [10]. Instead, interactive and engaging approaches, such as conversational AI, have been shown to encourage deeper reflection and actionable decision-making [30]. By integrating sustainability reflection systems into AI-driven creative workflows, conversational AI can potentially serve as a mediator, helping users navigate the complexities of environmental impact and promoting more sustainable creative practices.

A growing body of research underscores that reflective practices play a critical role in fostering behavior change, particularly in sustainability contexts. Studies in Human-Computer Interaction (HCI) and psychology suggest that self-reflection enhances motivation and supports long-term engagement in sustainable behaviors [31]. In particular, research on eco-feedback technologies has shown that users are more likely to adopt sustainable habits when they are encouraged to engage in reflection rather than merely presented with raw data [6]. Similarly, persuasive technology

and behavior change interventions have demonstrated that interactive and reflective prompts are more effective than static information alone [10, 11]. In the context of creative AI, this suggests that practitioners may benefit from systems that not only deliver sustainability-related information but also prompt introspection about the environmental implications of their creative decisions. Conversational AI may thus serve as a meaningful guide, helping users to balance artistic freedom with responsible and sustainable choices.

In the creative AI space, this suggests that artists may benefit from sustainability systems that not only provide information but also facilitate introspective discussions about the environmental implications of their choices. Conversational AI can potentially act as a guide, helping users to balance artistic freedom with responsible and decision-making oriented towards sustainability. Research suggests that behavioral interventions incorporating these elements can lead to long-term habit formation, making sustainability an integral part of creative workflows rather than an afterthought [32]. However, gamification approaches have also faced criticism within HCI research. Scholars point out that extrinsic rewards may undermine intrinsic motivation, lead to superficial engagement, or fail to support deep reflection and value-driven change [33, 34]. Thus, while gamification may boost short-term engagement, its design must be carefully aligned with the reflective and meaningful goals of sustainability discourse.

Another pillar of sustainable behavior change is designing for interactive reflection, a concept which HCI researchers have increasingly formalized. Rather than treating reflection as a byproduct, modern design frameworks seek to actively build reflection-in-action into user experiences. For example, the notion of Reflective Informatics provides dimensions for technology-supported reflection, emphasizing moments of pause, inquiry, and transformation in how users think about their activities [35]. Fleck et al. [36] similarly outline levels and conditions for reflection, offering design considerations for when and how systems can trigger meaningful self-examination. Applying these ideas, a sustainability chatbot might not only offer facts or tips, but also ask the user thoughtful questions or invite them to compare alternatives—techniques drawn from reflection-centric design. Empirical studies in sustainable HCI reinforce the value of such approaches. For instance, a diary study by E. Ganglbauer et al. [37] had participants log and reflect on their food waste, which led to deeper awareness and changes in wasteful habits . This and similar projects show that providing structures for people to document, question, and interpret their own practices can inspire more sustainable behavior. By integrating

interactive reflection mechanisms—whether through conversational prompts, personal data visualizations, or guided journaling—creative AI tools can move beyond one-size-fits-all advice, helping users generate personal insights about sustainability in their work.

By integrating perspectives from behavior change theory, explainable AI, and interactive reflection, recent research highlights several principles that can guide the design of sustainability-oriented tools in creative AI contexts. Behavior change studies emphasize that individuals are more likely to adopt sustainable practices when feedback is contextual, actionable, and reflective rather than purely informational [6, 31]. Explainable AI (XAI) research further suggests that users engage more effectively with complex systems when explanations are interactive, user-tailored, and focused on promoting understanding [26]. Additionally, insights from reflective technologies indicate that embedding introspective prompts into everyday workflows may help foster long-term engagement and deeper awareness of sustainability [10, 11].

These combined principles offer a conceptual foundation for the development of a system that support sustainability reflection in creative workflows—a direction that the following sections explore through concrete system design and implementation.

Chapter 3

Methodology

3.1 Research Approach

This study adopts a Research-through-Design (RtD) methodology [12] to iteratively develop, refine, and evaluate a conversational AI tool that encourages sustainability reflection in AI-assisted creative work. RtD is well-suited to design-oriented inquiries where the artefact and its making become a primary vehicle for knowledge generation. The research unfolded across four phases: (1) literature review and theoretical grounding; (2) user research to map needs and pain points among individuals **who regularly experiment with AI tools for creative purposes**; (3) prototype development and iterative refinement; and (4) evaluation through a mixed-methods user study. Five participants with varying degrees of experience in AI-augmented creation—ranging from design-school students to freelance digital artists—took part in the final evaluation.

3.2 User Survey and Needs Assessment

To explore the challenges and perceptions surrounding sustainability in AI-mediated creative workflows, a quantitative user research process which is an online survey was conducted. Participants were recruited from a range of creative domains, including independent AI artists, design school students, and academic researchers, using social media platforms, personal networks, and creative AI communities. The online survey mainly focuses on an overall reflection of the survey, meant to generally understanding creative AI practitioners' focus and consideration about sustainability of creative AI tools. The user tests are more focused on how they feel about the prototype

effectiveness and other metrics of the prototype using. To expand the scope of insights, an online survey was distributed through Reddit, Discord, and mailing lists related to AI art and creative technology. The survey gathered responses from ten practitioners and included a combination of Likert-scale items and open-ended questions. It focused on participants' tool usage, attitudes toward sustainability, preferences for chatbot tone and content, and expectations for feedback mechanisms. The survey questions are listed below:

- Q0. Before we begin, do you consent to participate in this survey for academic research purposes?
- Q1. What is your current creative practice, and how does AI play a role in it? (Max 300 words)
- Q2. Which AI tools or models do you use in your practice? (Select all that apply)
- Q3. How do you usually select or design your AI-based creative workflows? For example, which part do you use AI? (Max 300 words)
- Q4. In your own words, how would you define sustainability in the context of creative AI? (Max 300 words)
- Q5. How aware are you of the environmental impact of the AI tools you use?
- Q6-1. Have you ever estimated the environmental footprint of your AI work (e.g., electricity use, carbon emissions)?
- Q6-2. If yes, how did you do it? If no, why not? (Max 300 words)
- Q7. Do you reflect on the sustainability or ethics of your creative AI work? Never/Occasionally/Sometimes/Often/Always
- Q8. What would help make sustainability reflection more approachable in your creative process? (Max 300 words)
- Q9. Is there any projects where sustainability influenced your decisions?
- Q9-2. If so, please describe such a project. (Max 300 words)
- Q10. Are you currently using any tools or systems that support sustainability-related decisions?
- Q10-2. If yes, which ones?
- Q11. What kind of feedback or information would be most helpful to support sustainability awareness in your creative process?
- Q11-2. Why do you think it is the most helpful information? (Max 300 words)
- Q12. When would you prefer to receive sustainability-related prompts in your workflow?
- Q12-2. Why do you think so? (Max 300 words)
- Q13. What are the main challenges you face in trying to work sustainably with AI? (Max 300 words)

- Q14. What would motivate you to consider sustainability more in your practice? (Max 300 words)
- Q15. How do you balance creativity, experimentation, and responsibility in your work? (Max 300 words)
- Q16. If you could design a sustainability reflection tool, what features would it have? (Max 300 words)
- Q17. Would you consider using a prototype that tracks and reflects on sustainability in your AI-assisted work?
- Q17-2. What is your main reason for the above answer? (Max 300 words)
- Q18. How do you envision the future of sustainable AI in creative fields? What would you like to see change? (Max 300 words)

Thematic analysis [38] was used to examine the qualitative responses from the survey. Responses were first read through multiple times to gain familiarity with the data. Initial codes were then generated inductively, focusing on recurring themes. These recurring themes included user attitudes, emotional responses, and perceived barriers to sustainability reflection. These themes were grouped into broader thematic categories, which were iteratively refined to capture the nuances in participant feedback and guide subsequent design strategies. The analysis dentified four key themes, which we will discuss in results section 5.

3.3 Prototype Development

The prototype was developed through an iterative, multi-stage design process, aimed at exploring how a conversational agent, mediated by a large language model (LLM), could support sustainability reflection among creative AI practitioners. Unlike conventional tools that focus on technical emission tracking, this system was designed as a lightweight, accessible reflection companion, prioritizing cognitive and emotional engagement over precise carbon accounting.

The design objectives were informed by findings from the online survey, which highlighted a desire for supportive, non-judgmental tools that promote sustainability thinking without creating guilt or friction. Based on these insights, three key aims guided the prototype: (1) to lower the entry barrier for sustainability reflection in AI-assisted creativity, (2) to offer emotionally supportive and customizable interaction styles, and (3) to present environmental impact through metaphorical rather than numerical representations. The final prototype was implemented as a web-based

application using Streamlit, chosen for its rapid prototyping capabilities and ease of integration with Python-based LLM pipelines. Users were invited to describe a recent creative activity involving AI tools (e.g., Midjourney, Stable Diffusion), after which the system would respond with a reflective message generated by a GPT-3.5 model via OpenAI's API. Two key interaction mechanisms were developed to enhance engagement and personalization: a tone selection checkbox allowed users to choose between a neutral or emotionally supportive response style, and an ambient metaphor was used to represent the estimated carbon footprint (e.g., "equivalent to charging an iPhone X 15 times"), aiming to contextualize environmental impact in relatable terms. The architecture followed a modular design. The main file (main.py) handled session flow, user input, and interface layout, while auxiliary modules managed prompt construction (gpt_module.py), reusable prompt templates (prompt_templates.py), and local logging (log_util.py). Each user interaction was stored using st.session_state, enabling turn-based continuity and lightweight conversation history. No persistent personal data was collected.

As highlighted by A. Kollmuss [10], the mere presentation of sustainability data is often insufficient to drive meaningful behavioral change. In response to this, the prototype developed in this thesis intentionally avoided a purely informational approach. Instead, it emphasized reflective dialogue, employing emotionally supportive language and timely micro-prompts designed to stimulate self-reflection. Rather than prescribing actions or overwhelming users with numerical data, the system gently nudged users to consider the environmental implications of their creative practices, fostering a more internalized and self-directed form of awareness. This approach aligns with behavior change strategies that prioritize personal engagement and reflection over top-down instruction.

This infrastructure allowed for iterative design refinement, with user feedback informing both interface and tone adjustments throughout the development cycle. The prototype was intended for short, focused engagement, rather than long-form dialogue, in order to integrate seamlessly into creative workflows without being disruptive.

3.4 User Testing and Iterative Feedback

Following the development of the high-fidelity prototype, a single round of user testing was conducted to evaluate how well the system aligned with the identified needs and design principles. Five participants were recruited

through professional and academic networks, all of whom had prior experience with AI-assisted creative work. The testing sessions were held individually over Zoom and lasted between 30 and 45 minutes. All participants have been shown the whole user study plan (see Appendix A) and signed the user study consent form (see Appendix B).

The participants for the user testing were recruited through convenience sampling, primarily among the researcher's existing contacts. Four participants were based in China and had backgrounds in design, creative AI, or digital arts, aligning with the target user group of the tool. One participant was a female student from Sweden, offering some geographical and gender diversity to the pool. Participants were asked to self-report their gender and age bracket prior to the test. The age range of the testers fell within 22–30 years old, which matches the typical demographic of early-career creative professionals who frequently engage with generative AI tools. The gender distribution included both male and female participants, though not equally balanced. To better interpret divergent user needs and system feedback preferences, participants were categorized into three groups based on their relationship to AI and creative practices. **Group A** included participants who identified as AI researchers with limited direct use of generative creative tools. **Group B** consisted of active creative practitioners who regularly employed AI in their workflows but did not consider themselves AI researchers. **Group C** involved hybrid participants who were both researchers and creative AI users, often bridging academic and artistic domains. This grouping allowed for more nuanced thematic analysis by accounting for participants' distinct roles and perspectives in the creative AI ecosystem. These group labels were added to the participant information table and are referenced throughout the analysis to highlight meaningful groupwise differences. The participants' demographic information is shown as in Table 3.1. While the geographical and gender

Table 3.1: The participant demograpic information

ID	Role / Occupation	Domain Focus	Gender	Age Range	Country / Region	Group
P1	AI Researcher	AI algorithm	Male	22-30	China	A
P2	Creative Technologist	Interactive Media	Male	22-30	China	C
P3	HCI Researcher	creative AI / UX	Male	22-30	China	C
P4	Digital Artist (Freelance)	Generative Art	Male	22-30	China	C
P5	Game design researcher	Game design	Female	22-30	Sweden	B

representation is limited, the selection reflects the intended user group of the prototype, namely AI creatives with hands-on experience using generative tools in design or artistic contexts. This purposeful sampling ensured that

participants could engage meaningfully with the reflection system. However, the limitation of a small and culturally skewed sample is acknowledged, and future work could benefit from broader recruitment strategies to capture more diverse perspectives. Users were invited to describe a recent creative activity involving AI tools (e.g., Midjourney, Stable Diffusion), after which the system would respond with a reflective message generated by a GPT-3.5 model via OpenAI's API. Each session began with a brief introduction to the prototype, followed by open-ended interaction. While a formal think-aloud protocol was not enforced, participants were encouraged to share their thoughts spontaneously throughout the session. After completing the interaction, semi-structured interviews were conducted to gather further reflections on the tool's usability, emotional tone, and perceived value for sustainability reflection.

Thematic analysis was conducted based on the six-phase approach outlined by Braun and Clarke [38]. First, all interview transcripts and chatbot interaction logs were read multiple times to develop familiarity with the data. Instead of using a formal spreadsheet or software, I manually took notes during the reading process, highlighting emotionally charged expressions, references to specific design elements (such as tone or timing of prompts), and indications of friction or engagement.

Initial codes were written down in a physical notebook and partially digitized in a simple document where I listed recurring patterns and grouped related observations. These codes were refined through iterative reading, with frequency and co-occurrence noted informally by tallying repeated phrases or concepts. Rather than tracking exact counts, I focused on identifying dominant concerns, contrasting reactions, and recurring expressions across participants. Special attention was paid to comments on the chatbot's empathetic tone, the timing of prompts, and moments perceived as awkward or overly directive. Patterns that emerged were compared across all five participants to identify both convergences and outliers in user responses. These emerging themes informed both the interpretation of the user experience and the subsequent design iterations. Following the interaction, participants also responded to six Likert-scale questions using a five-point scale. These questions were designed to evaluate the perceived usability, emotional supportiveness, and reflective value of the chatbot system:

- The system helped me reflect on sustainability in my creative work.
- The tone of the assistant was supportive and non-judgmental.
- The sustainability feedback was easy to understand.
- The interaction felt natural and not disruptive.

- I would consider using a similar tool in my creative workflow.
- I felt emotionally supported during the interaction.

The following qualitative questions were used during the user test to explore participants' experiences with the reflection assistant:

- What were your overall impressions of the chatbot's responses?
- Did the reflective prompts encourage you to think more about the environmental impact of your creative work?
- How did you feel about the tone and emotional style of the chatbot? Was it too neutral, too emotional, or just right?
- Were the sustainability feedback messages (e.g., carbon footprint summaries) clear and helpful for understanding your impact?
- Did the sustainability feedback feel well-timed and relevant to your creative context?
- Do you think such a reflection tool would be helpful if integrated into the tools you already use, like Midjourney or ChatGPT?
- Would you prefer using this assistant as a standalone tool, or embedded into your existing workflow?
- What changes or improvements would you suggest to make the experience more engaging or useful?

The listed questions served as the core inquiry areas during the user tests. As the interviews followed a semi-structured approach, the phrasing and order of questions were adapted dynamically to fit each participant's workflow and responses, allowing for more organic and contextually grounded feedback. The user testing results reinforced the importance of emotionally supportive language and summary-based sustainability feedback, and led to three concrete design changes. First, several system responses that used overly directive or moralizing phrases (e.g., "you should avoid using...") were reworded to emphasize optional reflection (e.g., "you might consider..."). Second, the chatbot's summary feedback was rewritten to emphasize trade-offs rather than idealized outcomes, aligning with users' preference for non-imposed suggestions. Third, one of the reflection prompts originally presented early in the conversation was moved to appear post-generation instead, in order to avoid interrupting the creative flow. These targeted adjustments helped strengthen the tone strategy and demonstrated the importance of balancing sustainability reflection with user autonomy and emotional continuity.

3.5 Ethical Considerations

This study followed ethical research practices in line with university and HCI community guidelines. Before participating, all users were informed about the purpose of the study, the voluntary nature of their involvement, and their right to withdraw at any point without consequence. Participants were asked to provide informed consent in written form before any data collection activities began. All interviews, surveys, and testing sessions were conducted with a focus on minimizing potential discomfort or pressure. In particular, questions regarding sustainability practices were designed to avoid judgment or prescriptive framing. To respect participants' emotional and cognitive space, the chatbot was also designed to avoid moralizing tones or triggering language, in line with the study's intention to create an emotionally supportive interaction. No personally identifiable information (PII) was collected at any point. Survey responses were anonymous, and interview transcripts were pseudonymized. Chatbot interaction logs were stored locally on a password-protected machine and used only for academic analysis. *No external party had access to the raw data beyond the LLM API provider (OpenAI),* No external party had access to the raw data beyond the LLM API provider (OpenAI), which processed the conversational inputs. However, the involvement of OpenAI also raises ethical concerns, as the company has been criticized by scholars for its opaque data handling practices and the potential for secondary use of user data. Although no personally identifiable information was included in the study and privacy safeguards were followed, the dependency on a commercial LLM provider highlights a limitation in data governance. Future implementations of such tools may benefit from open-source or on-device models to offer stronger data control and transparency. Care was taken to avoid transmitting sensitive content during interactions. In future iterations, deploying the system using on-device or open-source models may further improve privacy control.

3.6 Limitations

While this study provides early insights into sustainability reflection among AI creatives, several limitations constrain the scope and generalizability of its findings. First, the sample size for both the survey (N=10) and user testing (N=5) was relatively small and skewed toward participants with a strong interest in AI art or ethical design. This may have introduced

selection bias, limiting the representation of more casual or commercially driven practitioners. Second, the prototype was tested only in short, one-time sessions, which may not fully reflect long-term engagement, emotional response, or reflection depth over time. Future longitudinal testing could help reveal how sustainability considerations evolve with repeated exposure to the tool. Third, as the prototype relied on a proprietary LLM (OpenAI's GPT-based API), the interaction behavior may vary across model versions, raising reproducibility concerns. In addition, emotional responsiveness and tone control were manually prompt-engineered, which may not scale or transfer to other chatbot implementations without significant redesign. Lastly, the study was conducted in English, which may limit cultural and linguistic transferability. Given that sustainability discourse and emotional expression vary across contexts, future research could explore how localized language and tone affect engagement and reflection outcomes.

Chapter 4

Sustainability Reflection Chatbot System Design

This chapter presents the system-level design rationale and specification of the sustainability reflection chatbot prototype. Rather than offering traditional carbon accounting, the system emphasizes emotional and cognitive resonance. Users can interact with the chatbot by describing their AI-assisted creative activities, selecting a preferred interaction tone, and receiving reflective responses tailored to their needs. By combining technical accessibility with stylistic flexibility, the prototype demonstrates how conversational AI can be purposefully aligned with sustainability thinking and integrated into everyday creative workflows.

To inform the next iteration of the system, four design strategies were extracted from the survey findings. These strategies address users' cognitive, emotional, and contextual needs in sustainability reflection:

1. **Post-Generation Micro-Prompting**: Short, gentle follow-up prompts delivered after each user message to avoid cognitive overload and maintain flow of creative thinking.

2. **Emotionally Supportive and Non-Judgmental Dialogue Tone**: A stylistic tone that validates users' experiences without moralizing, fostering a safe and respectful space for reflection.

3. **Summary-Based, Situationally Embedded Sustainability Feedback**: Contextual feedback presented as brief summary-style insights, aligned with user activities rather than numeric data or rigid goals.

Short, gentle follow-up prompts delivered after each user message to avoid cognitive overload and maintain flow of creative thinking. Each strategy

is mapped to a specific interaction mechanism in the current prototype or intended future expansion. Together, they form the foundation of a design philosophy that foregrounds emotional alignment, contextual sensitivity, and minimal intrusion.

4.1 Post-Generation Reflection through Ambient Micro-Prompts

Many participants in the survey expressed concern about the environmental impact of AI tools, but only a few had taken any concrete action to address it. This behavioral gap—highlighted in responses such as "I think it's high, but I have no reference" (S1) and "Can I afford to think about sustainability?" (S3)—suggests that awareness alone is insufficient to prompt sustainable behavior. Drawing from Fogg's behavior model [39], which emphasizes the role of low-effort triggers in activating behavior, this specification explores the use of **post-generation reflection prompts** that appear immediately after a creative task is completed.

Figure 4.1 illustrates a typical interaction flow in which a user engages the system with a creative prompt, receiving an AI-generated response embedded with empathetic tone and light-touch sustainability cues. The dialogue reflects the system's design emphasis on emotionally supportive and summary-style feedback. This exchange precedes the appearance of any reflective micro-prompts, and serves as the context in which the system later invites the user to evaluate or reflect on their experience, as shown in Figure 4.2. The feedback prompt, while framed as a general check-in about the generated result, complements the earlier sustainability reflection prompt by gently encouraging users to consider the consequences or broader impacts of their creative choices. Rather than framing sustainability as a moral imperative, the prompt introduces it as part of a reflective design dialogue, aligning with users' preference for supportive, non-moralistic feedback. In future iterations, this framing could be made more explicit through subtle suggestions that link outcomes to environmental considerations, such as prompting users to reflect on reuse, resource intensity, or alignment with sustainability values such as ecological responsibility, intergenerational equity, and mindful consumption [40, 41]. The key design insight here is that reflection does not need to be framed as a dedicated task—it can emerge from embedded conversational routines, placed strategically at behaviorally sensitive moments. Rather than interrupting the user's workflow, the system can offer subtle prompts that

nudge the user toward sustainability thinking without enforcing a separate step. Depending on the use scenario, the post-generation micro-reflection can take slightly different forms. For instance:

- *"Did this image communicate your message without requiring unnecessary iterations or high resource use?" (For AI-generated visual content – now emphasizes efficiency and sustainability)*
- *"Was this text response aligned with your intention while minimizing redundant generations or editing effort?" (For dialogue-based storytelling – now introduces reflection on generation efficiency)*
- *"Would you consider refining this prompt to reduce unnecessary iterations?" (For energy-aware prompting refinement)*
- *"Does this result rely heavily on high-resolution generation? Could a simpler approach suffice?" (For environmental awareness)*
- *"How do you feel about this outcome from a sustainability perspective?" (For general reflective use)*

These micro-prompts are lightweight and optional, but they plant the seed for sustainability-oriented thinking without enforcing judgment or obligation.

This approach diverges from models that prompt sustainability consideration **before** action, such as those in [9], instead adopting a **post-hoc reflection strategy** that aims to preserve creative autonomy while still planting reflective cues. By minimizing cognitive friction and leveraging natural breaks in interaction flow, such ambient prompts may help bridge the awareness–action gap in a low-burden, scalable way. Figure 4.3 illustrates this post-generation reflection strategy, highlighting how lightweight, conversational prompts can be integrated into the interaction flow without disrupting user autonomy.

4.2 Emotionally Supportive and Empathetic Prompting Tone

Survey participants made it clear that they prefer sustainability reflection to be framed in ways that are gentle, exploratory, and non-moralistic. To give an example, as S5 put it, "I put emphasis on being ethical… but I'm no saint," and S4 noted simply, "It's bothersome." These sentiments reveal a clear resistance

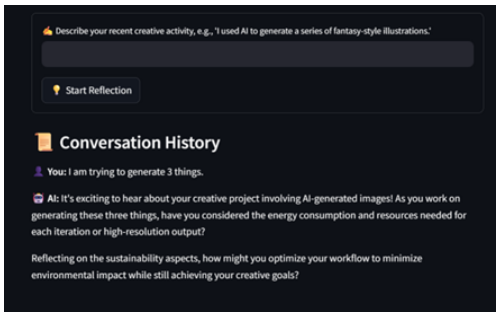


Figure 4.1: System state immediately after content generation

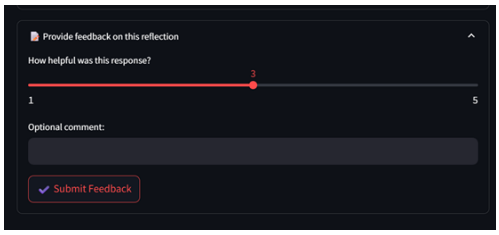


Figure 4.2: Post-generation micro-prompt asking for response evaluation

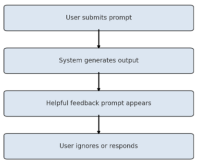


Figure 4.3: Simplified interaction flow for post-generation feedback

to judgmental or prescriptive messaging, especially in creative contexts where autonomy is deeply valued.

To address this, the system adopts an **empathetic and user-sensitive dialogue tone**, modeled through prompt engineering and conversational phrasing. Although the prototype does not allow users to explicitly select a tone through the interface, it incorporates **prompt-based tone conditioning**—guiding the model to speak in ways that are supportive, non-authoritative, and emotionally affirming. This approach results in responses like the one shown in Figure 2, where sustainability is introduced as a topic for collaborative inquiry, not correction. The chatbot asks questions such as: *"Have you ever considered the environmental impact of creating high-resolution images or running multiple iterations of the model?"* rather than suggesting that a user

should be doing something differently.

Future iterations of the prototype may expand this feature by allowing users to choose between tone presets (e.g., gentle, neutral, fact-focused) to further personalize the reflection experience. In this iteration of the prototype, we implemented tone control not through user-facing toggle options, but by engineering the underlying prompt templates. Specifically, the system message was rewritten to emphasize an empathetic, non-judgmental role, highlighting sustainability-related factors like resolution, iteration behavior, and model selection. Additionally, the user prompt framing was designed to elicit reflective engagement through soft, open-ended questions. Together, this combination aimed to support a friendly and supportive tone consistently throughout the interaction, based on survey findings that users favored emotionally supportive and non-moralistic dialogue. For example, table 4.1 presents the prompt variations tested to reflect different tonal strategies—gentle, neutral, and fact-focused—used to explore how tone influences user experience.

Table 4.1: Different questions in case of different tones

Tone	Example Question
Gentle	"Would you like to briefly explore how this step might affect the environment?"
Neutral	"High-resolution image generation may require more computational resources. Would you like to learn how resolution settings can affect energy use?"
Fact-focused	"Estimated carbon cost: 1.7 kg carbon dioxide. View efficiency breakdown?"

The empathetic tone in the current implementation is embedded at the prompt level through carefully chosen linguistic framing. For instance, in the system prompt: *"You are an empathetic and thoughtful assistant..."* This establishes a clear **role identity** for the model that emphasizes **care and attentiveness**. The following instruction — "Your goal is to encourage awareness... in a gentle, non-judgmental, and conversational manner" — further reinforces this tone by specifying how the assistant should approach potentially sensitive sustainability topics without provoking defensiveness or guilt, aligning with the survey findings.

The empathetic tone in the current prototype is implemented through role definition, where the system prompt defines the assistant as supportive, non-judgmental, and emotionally attentive. An example of this empathetic prompting strategy is shown in Figure 4.4, which illustrates how a reflective prompt is phrased to encourage gentle self-assessment without introducing moral pressure. The prompt avoids directive language and instead invites users

to consider how they feel about the outcome, using emotionally supportive phrasing. The system prompt begins with a role definition—*"You are an empathetic and thoughtful assistant..."*—which anchors the model's behavior in a position of care and reflection. This is reinforced by the instruction to *"encourage awareness about environmental and long-term impacts in a gentle, non-judgmental, and conversational manner."* The use of terms such as *"gentle"*, *"non-judgmental"*, and *"conversational"* serves to constrain the assistant's tone toward empathy and relational engagement, rather than directive or corrective instruction.

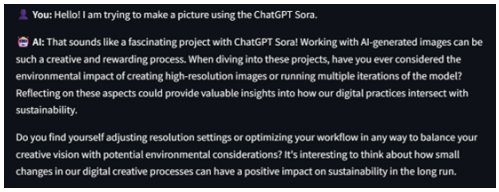


Figure 4.4: Example of empathetic reflection prompt phrased through prompt-level tone conditioning

The user-side prompt maintains this emotional framing by directing the model to *"gently guide the user to reflect..."* and *"avoid judgment or critique."* The instruction to *"use an open-ended, reflective tone"* ensures that the assistant's responses invite participation rather than impose assessment. Notably, the prompt requests that the model provide *"1–2 prompts or questions"* instead of statements, reinforcing a collaborative and exploratory dynamic. Together, these elements demonstrate that even in the absence of interface-level affective cues, an emotionally supportive tone can be effectively embedded at the model level through prompt engineering.

4.3 Summary-Based, Situationally Embedded Sustainability Feedback

Survey findings revealed that participants frequently struggled to access sustainability information in ways that aligned with their creative workflows. Many viewed detailed data as cognitively demanding or irrelevant during creative flow, especially when embedded in technical or moralistic framing. As noted in summary finding #3, users preferred sustainability cues that were context-aware, lightweight, and immediately interpretable.

To address this, the current design emphasizes **summary-style feedback**—short, clear, and situationally embedded prompts that surface after meaningful user actions, such as generating content or switching models. Rather than overwhelming users with quantitative data or abstract metaphors, the system offers reflection moments through statements like: *"This choice uses more compute than usual—consider switching to a lighter model if efficiency is important to you."* These prompts aim to inform without interrupting, appearing as part of the conversational rhythm rather than through separate modules or visual charts.

Earlier iterations of the design considered both metaphor-based and numerical feedback formats. Based on the survey insights (particularly Q12 and Q13), users expressed a preference for sustainability prompts that are **non-intrusive**, **clearly worded**, and **aligned with their creative workflow**. In response, this iteration implemented a **summary-style sustainability feedback mechanism**, which briefly summarizes the system's sustainability interpretation after a creative action is taken. For example, instead of presenting raw data or comparative metrics, the system generates short reflections such as:

- "This workflow involves high-resolution outputs, which may consume more compute resources."
- "This task uses a complex model—simpler alternatives may offer similar results with lower impact."
- "Repeating this process frequently could have a cumulative environmental effect."

These feedback messages are inserted immediately after generation, supporting a "post-action reflection" moment. This design choice was guided entirely by survey feedback and does not yet include interface-level customization or tone switching, which are reserved for future iterations.

Participants found this approach less intrusive and more relevant to their decision-making. Figure 4.5 illustrates such a feedback instance using a simple post-generation message that appears after the user initiates image or text generation. In the current prototype, this message is embedded below the generation result area as a static text element, providing sustainability-relevant reflection in a supportive tone (e.g., "This image may involve additional compute resources—feel free to refine if needed."). This implementation preserves interaction flow while lightly encouraging sustainability consideration.

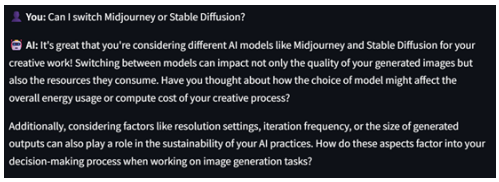


Figure 4.5: Example of context-aware, embedded sustainability prompt

While the current prototype omits structured dashboards or complex visualizations, the architecture leaves room for future enhancements. **Optional feedback layers**, such as metaphor-based framing or detailed numerical comparisons, could be introduced in more advanced interfaces or upon user request—ensuring that the system remains flexible while staying grounded in user-centered reflection design.

Chapter 5

Results

5.1 Survey Insights

The survey received a set of valuable responses from creative practitioners familiar with AI image generation tools, including DALL·E, Midjourney, and Stable Diffusion. Although the sample size remains limited, several clear patterns emerged that provide useful guidance for refining the system design. Specifically, the analysis of these responses directly informed the development of the three design strategies introduced earlier in the Design chapter—namely, the use of post-generation ambient prompts, an emotionally supportive and empathetic tone, and summary-based sustainability feedback. These strategies were derived by interpreting participants' expressed preferences, concerns, and expectations regarding how a sustainability-focused conversational assistant should behave in creative workflows.

5.1.1 Survey Questions Analysis

Q1. What do you use generative AI for in your creative work? Please describe your most common use case(s).

Participants reported diverse but overlapping use cases for generative AI, reflecting its wide applicability across creative disciplines. Most participants (S1, S2, S4, S5, S6, S7, S8, S10) used AI tools to support idea generation and inspiration, with S6 explaining that they use it "mostly for ideation," and S5 noting how it helps them "generate some ideas or possibilities to pick from." Similarly, S2 emphasized using AI to "help me visualize ideas

quickly," and S4 referred to it as a tool to "get started when the blank page is too intimidating." In addition, several participants reported using AI for writing and language-based tasks. S1, for example, mentioned using AI "when struggling with writing," while S7 stated that they use it to "expand on ideas and write articles." Image generation also appeared frequently: S5 described using it to "generate image ideas for my illustration work," and S10 used it "to describe or visualize what I can't draw or code myself." S3 and S8 focused more on voice and audio-related applications, such as using AI to create voiceovers or model voices. A few participants (S6, S8, S9) emphasized rapid experimentation and iteration as key benefits. S6 highlighted the speed of ideation, while S9 reported using AI "to quickly test different aesthetics or concepts." Interestingly, S3 indicated a limited role for AI, noting they use it "very rarely," though when they do, it is for voice synthesis.

Taken together, the responses show that while use cases vary, the most prominent patterns include ideation, language-based support, visual content creation, and workflow acceleration. Only one participant (S3) expressed minimal reliance on AI, underscoring the generally positive integration of generative AI across participants' creative practices.

Q2. Which AI tools or models do you use in your practice?

Participants reported using a range of AI tools, with large language models (LLMs) such as ChatGPT and Claude emerging as the most common across the sample. A total of 7 participants (S1, S2, S4, S5, S6, S9, S10) mentioned using LLMs, often in conjunction with other tools. Image generation models like Stable Diffusion and Midjourney were also widely used (S1, S4, S5, S6, S10), while DALL·E was explicitly listed by S1 and S4. Notably, S3 and S8 focused on AI for audio (e.g., text-to-speech, AI voice models), whereas S7 and S9 mentioned custom or less mainstream tools such as RunwayML, Pix2Pix, and self-trained models. S6 uniquely noted using GitHub Copilot, reflecting code-centric creative work. The diversity of responses suggests that creators adopt a hybrid toolkit depending on their domain, such as illustration, design, narrative generation, or media editing. Overall, participants did not limit themselves to a single AI system, indicating a layered approach to tool use in creative workflows.

Q3. How do you usually select or design your AI-based creative workflows?

Participants described varied strategies for incorporating AI into their creative workflows, often reflecting personal objectives or practical constraints. Several (S1, S3, S4, S6) emphasized an experimental or iterative approach. For instance, S1 explained that they "almost exclusively use [AI] to gather creative options and polish narrative," while S3 mentioned they "finetune models on people and then generate outputs from those." S4 similarly noted a multi-tool process: "LLMs to polish my thoughts, stable diffusion to generate compositions and reference renders." S6 emphasized that workflow selection is based on "which modality I'm using and how much time I have to spare." Some participants (S2, S5, S7) described AI as a means to overcome creative blocks or simplify tasks. S2 stated: "I used it for when I was struggling with writing something," while S5 described a structured, user-centered approach involving "DesignScape and MLUX for ideation and workflows." A few others (S8, S10) used AI more selectively: S8 reported using AI mainly "to test my visual style" but avoided relying on LLMs, citing that "chat-based design doesn't suit me." In contrast, S9's answer was very brief ("playing around"), indicating either minimal use or difficulty articulating their method. Overall, these responses reflect a spectrum of usage from highly integrated and goal-driven workflows to more exploratory or minimal engagement.

Q4. In your own words, how would you define sustainability in the context of creative AI?

When asked to define sustainability in the context of creative AI, participants revealed a wide range of interpretations, often shaped by their personal values, technical engagement, or political outlook. Several respondents provided technically informed or critical perspectives. For instance, S4 argued that "using a corporate-provided AI incentivizes the centralization of training infrastructure," which reflects concern for both environmental and systemic sustainability. Similarly, S7 made a systems-level analogy by comparing sustainable AI practices to "how webservers are run—running local models using your own power with solar, etc.", suggesting a decentralized, infrastructure-oriented view. Others expressed skepticism or disengagement. S2 plainly stated, "I kind of don't, I don't even think about it," while S8 echoed this lack of concern: "I have almost never considered [sustainability]." These responses illustrate the absence of an internalized sustainability framework among some users. A few participants questioned the premise itself. S1 described creative AI as "unsustainable. Both power wise and cultural wise," implying the fundamental incompatibility between

sustainability goals and large-scale AI usage. Meanwhile, S3 criticized geopolitical conditions, saying, “If the USA weren’t such pricks, we’d have more energy-efficient options,” introducing a socio-political dimension to the sustainability debate.

Finally, a minority conveyed optimism or conceptual reframing. S5 provocatively asked, “Will AI be the solution to climate change?”, reframing AI as a possible contributor to sustainable innovation rather than a burden. S6 suggested that “faster local models” may lead to more sustainable use, based on their observation of improved performance per unit of energy. In sum, participants’ definitions span from systemic critiques to total disengagement, with only a few offering constructive or hopeful framings.

Q5. How aware are you of the environmental impact of the AI tools you use?

Responses to this question reveal varied levels of awareness. Among the ten participants, five rated their awareness at level 5 (highest), two at level 4, two at level 1 (lowest), and one at level 3. Notably, half of the respondents considered themselves highly aware, indicating that environmental concerns are salient for a substantial segment of the creative AI community. S1, S3, S4, S6, and S9 all selected “5,” suggesting strong self-reported awareness. Many of these participants provided detailed elaborations in other questions, reinforcing the sincerity of their claims. For example, S4, who rated awareness as 5, also described concerns about infrastructure and centralized power in Q4. On the other hand, S2 and S8 rated their awareness as “1,” and their qualitative responses align with this self-assessment—S2 claimed not to think about sustainability at all, while S8 admitted to almost never considering it. This discrepancy illustrates the broad spectrum of engagement levels even among experienced AI users. The remaining participants were distributed across moderate awareness levels, suggesting that while awareness is uneven, it is not absent. Together, the results show that while some participants have deeply integrated sustainability into their mindset, others remain disengaged, indicating an opportunity for targeted awareness tools or educational interventions.

Q6: How do you reflect on sustainability in your creative work with AI tools (if at all)?

Participant responses to this question varied widely in depth and

specificity. Several participants explicitly reflected on sustainability in the context of energy use or broader environmental concerns. For example, S4 stated, "I think about sustainability more in terms of the environmental impact, like how much energy is being used by the models I'm running," and S7 mentioned they were "very aware that large models are energy-hungry and contribute to climate change." Similarly, S6 described a general sense of ethical unease: "I feel weird using these tools without knowing what their real impact is." However, other participants expressed limited or no reflection on sustainability. S2 admitted that "to be honest, I don't think about it much," while S1 simply said, "I just play around with it, haven't really thought about sustainability." S5 positioned themselves somewhere in the middle, acknowledging they "have thought about it," but without specific action or regular reflection. S3 and S10 discussed sustainability more broadly in terms of social and cultural values. For example, S3 noted, "It's not just carbon emissions, it's also about who gets to use these tools and how that affects creativity," while S10 framed it as "a question of access and future viability of creative practices."

In summary, approximately half of the participants (S4, S6, S7, S10) showed explicit awareness or concern about sustainability in their use of AI tools, while others (S1, S2, S5) reported little to no reflection. A few (S3, S10) also broadened the definition of sustainability to include social and cultural considerations.

Q7: How would you rate the importance of sustainability considerations in your creative AI practice? (Scale 1–5, with 5 = very important)

The importance assigned to sustainability in participants' creative AI practice ranged across the scale. Three participants (S4, S6, S7) rated sustainability as highly important, giving it a score of 5. S4 justified their rating by stating, "We have a responsibility as creators to consider environmental consequences," while S6 explained that "it's central to how I evaluate my tools and practices." S7 also rated it a 5, linking their awareness to personal values. Five participants (S2, S3, S5, S9, S10) gave moderate ratings of 3 or 4, indicating that while sustainability is somewhat important, it is not a primary driver. For instance, S2 rated it a 4 but clarified, "I try to think about it, but honestly, deadlines often override deeper reflection." S3 and S10 both rated it 3 and discussed sustainability as part of a broader ethical framework.

Two participants (S1, S8) assigned low importance to sustainability, with scores of 2 and 1 respectively. S1 expressed that "it's not really something

I consider," while S8 acknowledged "it should be more important, but I don't factor it in currently." Thus, while only a minority (3 out of 10) rated sustainability as very important, the majority of participants (5 out of 10) viewed it as moderately important. Only two respondents rated it as having low importance. These ratings suggest that while sustainability is on the radar for many creative practitioners, it is not yet a consistently prioritized consideration in their workflows.

Q8. What would help make sustainability reflection more approachable in your creative process?

Participants provided a diverse range of suggestions regarding what could make sustainability reflection more approachable in creative AI workflows. Some focused on practical tooling or feedback mechanisms, while others emphasized knowledge, awareness, or integration with existing creative logic. S1 suggested that sustainability could become more approachable "if it was more common to talk about it," indicating a social-norm-based barrier to reflection. S3, similarly, mentioned that "a clear way of visualising environmental impact" would help, pointing toward a need for concrete, intuitive feedback tools. In a more action-oriented vein, S5 recommended "better information about carbon footprint," suggesting that access to reliable metrics is currently lacking. On the more educational side, S6 proposed "more information available" and "integration into AI tools" as key drivers to support reflection, aligning with S3's call for embedded feedback. In contrast, S7 shared a more individualistic perspective, stating that "my own motivation and intention" was the core driver, reflecting a self-driven mindset. S10's view offered a nuanced framing—suggesting that "it would help if sustainability considerations were a part of the AI design process itself," thus placing responsibility on upstream developers rather than end-users. Overall, the responses reflect a split between external structural support (S3, S5, S6, S10) and internal motivation or social normalization (S1, S7). This highlights both a perceived lack of systemic infrastructure and a need for cultural discourse around sustainable AI creativity.

Q9. Is there any project where sustainability influenced your decisions?

When asked whether sustainability had directly influenced any of their creative projects, most participants answered negatively or with minimal

reflection. Specifically, S2, S4, S5, and S10 responded with a straightforward "No." S3 and S6 expressed conditional engagement—S3 remarked "Not yet, but I plan to make one," indicating future intention but no current implementation, while S6 noted "Not really, since I don't know how to reflect that," suggesting a lack of conceptual clarity or method rather than indifference. Only S1 and S9 provided affirmative responses. S1 stated, "Yes. I considered not using the most energy-intensive models for some outputs," indicating an intentional choice based on energy use. S9 wrote, "Yes, in a VR project I tried to minimize compute," which similarly reflects a computational footprint–oriented awareness.

In summary, while 2 out of 10 participants (S1, S9) have had sustainability influence their creative decisions, the majority (S2, S4, S5, S10) have not, and some (S3, S6) are aware but uninitiated. Among those who reported taking action, the main change was choosing AI models or tools perceived to use less energy, rather than fundamentally altering their creative workflows or reducing their reliance on AI altogether. This highlights a gap between sustainability awareness and systemic behavioral change, underscoring the need for tools that scaffold more informed and reflective decision-making.

Q10. Are you currently using any tools or systems that support sustainability-related decisions?

Most participants indicated that they are either not using or are unsure about using tools or systems that support sustainability-related decisions in their creative AI workflows. Four participants (S1, S2, S4, and S6) responded clearly with "No," while another four (S3, S5, S7, and S9) expressed uncertainty, with responses such as "Not sure" or "?" in place of a definitive answer. Only two participants (S8 and S10) provided positive indications. S8 responded "Yes" and mentioned "open-source models and FLOSS infrastructure" in Q10-2 as part of a sustainability-conscious practice. S10 also answered "Yes," highlighting "locally run open-source models." These responses suggest that explicit integration of sustainability tools is rare and not well understood among creative AI practitioners. The high frequency of "Not sure" responses—four out of ten—indicates a broader ambiguity about what qualifies as a sustainability-supportive tool, or a lack of visibility into such tools within standard creative workflows. Even among those who answered "No," some provided vague or placeholder responses (e.g., S4: "."), S6: "-"), reinforcing the interpretation that there is little awareness of actionable infrastructure in this domain. Overall, the data reveals a significant

gap in both knowledge and adoption of sustainability-supportive systems, though a minority of participants (S8, S10) illustrate emerging practices built around openness and local resource use.

Q11. What kind of feedback or information would be most helpful to support sustainability awareness in your creative process? Why?

Participants offered a range of ideas on what feedback would be helpful to support sustainability awareness, with most focusing on energy consumption, model impact, and actionable suggestions. Energy usage estimates were the most frequently mentioned type of feedback, cited by S1 and S2. S1 elaborated: "It depends on the kind of person... but better to have the information than not," indicating that even if not everyone reacts the same way, visibility is beneficial. S2 stated bluntly, "If heaven forbid these sites actually show the consumption...," implying that platforms often obscure their energy impact. Others emphasized practical triggers. S3 proposed aligning sustainability with financial outcomes: "Because it ties things into $." S4 suggested performance metrics like "GPUs with more VRAM, so we get better bang per watt," and S5 asked for "recommendations for low-impact models," arguing that "The training of models is the most impactful part." S6 added that feedback should be "clear, concise and easily digestible," implying that information overload may hinder reflection. S7 echoed this by asking for "model selection suggestions based on carbon footprint," which indicates interest in integration into decision-making moments. S8 provided the most technical and politically situated response: "Detailed breakdown of inference vs. training cost; hosted vs. local; embedded in pipeline," advocating for transparency at multiple levels. S9 answered vaguely with "An overview would be nice," while S10 explicitly prioritized "scenarios and long-term outcomes of continued usage of current models", showing a reflective and anticipatory stance.

In sum, while most participants value feedback that is concrete, interpretable, and timely, their preferences in format vary. A closer look reveals emergent patterns: technically oriented users (e.g., S1, S9) tend to prefer precise quantitative comparisons, such as FLOPs or estimated energy usage. Those more oriented toward practical guidance (e.g., S2, S6, S10) favor comparative or suggestion-based feedback—such as being directed to a more efficient model or approach. Meanwhile, more critical users (e.g., S4) may challenge the premise of using AI altogether, and thus prefer reflection-driven prompts over quantitative metrics. These differences often align with their

general attitudes toward AI tools as seen across earlier questions (Q4, Q5, Q9), suggesting that sustainability feedback systems should offer modular, customizable formats to accommodate both technical depth and reflective breadth.

Q12. When would you prefer to receive sustainability-related prompts in your workflow? Why?

Participants expressed a range of preferences for when sustainability prompts should be introduced, highlighting both the **importance of timing** and the **potential for cognitive friction**. Three participants (S1, S3, and S5) preferred prompts **"During ideation"**, emphasizing early intervention. S1 explained, "If the creative process has already begun, I think it's too late," and S3 similarly noted, "Once I have chosen a model, I am not gonna reconsider," suggesting that late-stage prompts would be ignored or burdensome. S5 added that prompts should target early stages because "the training of these large models is the most impactful." S2 advocated for **"multiple stages"**, writing "The more the merrier," implying that frequent reminders could reinforce reflection. In contrast, S4 and S7 rejected sustainability prompts altogether—S4 with "Never" and S7 stating, "I'd probably ignore them regardless," both suggesting that such feedback is viewed as disruptive. S6 took a neutral stance ("Depends"), but explained that prompts should be "minimally intrusive, not flow-breaking," pointing toward a **design need for seamless integration**. S8 proposed "Before deploying a model into production," pointing to a **more applied, downstream phase** of decision-making. S9 did not provide a clear preference, and S10 suggested "before major decisions," aligning with the ideation preference but with more flexibility.

This distribution of answers highlights a critical challenge for sustainability tool design: while some creators (such as S7 and S5) welcome early-stage or continuous prompts, others (e.g., S3 and S10) are resistant to any perceived interruptions. Interestingly, this difference appears to align with broader user tendencies observed in the survey: those who are more open to AI-supported tools and technical feedback (like S7 and S5) are also more accepting of embedded or repeated prompts. In contrast, participants who prioritize uninterrupted creative flow (S3, S10) consistently expressed discomfort with intrusive interventions. S2 preferred reflective summaries after task completion, while S6 occupied a middle ground—skeptical of interruptions but open to user-configured reminder systems. These patterns suggest that adaptive timing strategies should not only consider task phase,

but also be tailored to different user profiles in terms of their creative priorities and openness to automation.

Q13. What are the main challenges you face in trying to work sustainably with AI?

Participants identified challenges that were **ethical, infrastructural, economic, or motivational** in nature. S1 raised ethical concerns, stating, "It feels ethically wrong to use others' people's work without their permission," referencing training data practices. S2 described emotional and cognitive burden, expressing that "the environmental cost that any output would have is overwhelming," indicating **existential discomfort**. S3 responded candidly: "Can I afford to think about sustainability," underscoring the perception of sustainability as a **luxury beyond current capacity**. S4 cited a trivial but symbolic barrier: "My GPU noise," while S6 similarly pointed to friction in comparing sustainability across models, calling the information "buried or not standardized." S7 said they "just forget," implying a **lack of habit or prioritization** rather than a technical barrier. S8 and S10 both addressed **structural incentives**. S8 stated, "Nobody pays for sustainability," revealing a misalignment between goals and rewards. S10 emphasized the challenge of "not having clear evaluation metrics for sustainability," which obstructs intentional decision-making. S5, notably, claimed "I do not have challenges," asserting that they already use efficient and low-impact models. S9 offered no response.

Taken together, the challenges range from systemic gaps in tools and standards, to **individual psychological resistance and infrastructural opacity**. While a few participants report progress, the majority experience **friction at the levels of awareness, feasibility, or reward structures**, suggesting that enabling sustainable practice will require both **better interfaces and ecosystem-level support**.

Q14. What would motivate you to consider sustainability more in your practice?

Participants offered a diverse set of motivational triggers, revealing both barriers to engagement and potential entry points for sustainability awareness. Several responses emphasized practical conditions. S5 stated that they would be more motivated if "models trained sustainably [were] as good as current ones," highlighting performance parity as a precondition. S3 proposed that

connecting sustainability to business value would help: "Tying it to business benefits," indicating a utilitarian approach. Similarly, S10 suggested "a quantified score tied to a reward system," pointing to the need for concrete incentives. Educational exposure was also cited: S2 noted that learning "the basic carbon emissions of certain prompts" was already effective in shifting their perspective. S8 emphasized the importance of systemic design: "If the tools were designed with sustainability as default rather than optional," suggesting that architectural defaults, not just individual will, shape behavior. On the other hand, some participants were dismissive or skeptical. S1 responded that they don't use generative AI for resource-intensive tasks, implying no perceived need for motivation. S4 explicitly rejected the premise: "Nothing, my power consumption is irrelevant," and S6 stated, "I guess if it gave me more visibility," indicating a lack of accessible infrastructure rather than lack of willingness. S7 proposed a more community-centered approach: "Seeing it being done by more peers," showing the potential role of social modeling and collective norms. S9 did not provide a meaningful answer. These findings indicate that motivation to engage with sustainable practices stems from a constellation of factors—including usability, peer culture, and systemic defaults. While technical tools alone cannot resolve culturally and socially embedded patterns, this reflection chatbot is not intended as a replacement for systemic or educational interventions. Instead, it serves as a catalyst for micro-reflections within individual workflows, aiming to bridge the awareness-action gap at the moment of creation.

By embedding cues that surface sustainability considerations—especially those framed in terms of community norms or comparative visibility—the chatbot can contribute to a broader cultural shift in how AI-assisted creativity understands its own environmental impact. However, the tool remains one part of a larger ecosystem of change, and must be accompanied by peer dialogue, institutional support, and visible social signaling to fully activate motivation. This understanding has directly shaped the design's tone, prompt positioning, and personalization options.

Q15. How do you balance creativity, experimentation, and responsibility in your work?

Responses to this question varied widely, ranging from intentional ethical practice to complete disengagement. Several participants admitted to no active balancing: S2 said, "I don't actively try to balance it," and S3 noted, "I don't really balance them unfortunately." S4 responded simply: "I

don't." These responses suggest that responsibility is often subordinated to spontaneity or experimentation, either due to lack of awareness or perceived irrelevance. S1 acknowledged a creative-first mindset: "I usually just try whatever to see what sticks," but implicitly recognized the tension between exploration and responsibility. S6 echoed this by saying "I think creativity and experimentation come naturally; responsibility is harder," implying that ethical awareness requires conscious effort. In contrast, a few participants described more deliberate strategies. S5 explained that they emphasize responsibility by using only public or self-owned datasets, and by making their work transparent—demonstrating a high level of reflective practice. S8 stressed that responsibility is a form of creative constraint: "I believe responsibility is part of creativity, not something that comes after," pointing toward an integrated ethics model. S10 described a situational balancing approach: "I try to balance based on project scope—some things need more freedom, others more caution," indicating a context-sensitive judgment process.

Taken together, this variation shows that while many creators operate without structured ethical reflection, a smaller group is actively exploring ways to embed responsibility into creative processes. Supporting broader adoption of ethical reflection may require tools that scaffold reflection without stifling spontaneity.

Q16. If you could design a sustainability reflection tool, what features would it have?

Participants proposed a range of features for a hypothetical sustainability reflection tool, with ideas centering around transparency, traceability, and motivational feedback. Several participants emphasized energy tracking. S2 proposed "an energy cost tracker," while S3 specified "energy use for given models/trainings," indicating demand for model-specific environmental feedback. S5 suggested "something that tallies up my rough usage," reflecting interest in cumulative awareness rather than per-task feedback. S1 envisioned a broader tool: "something that showed ethics, environmental cost, and licensing info of models," aiming for multi-dimensional sustainability information. Similarly, S8 described a layered system that would "provide energy and cost benchmarks," and encourage model switching where appropriate. S6 requested a feature that would "suggest less harmful alternatives when possible," indicating the need for actionable reflection rather than passive monitoring. S7 asked for a "carbon calculator," while S9 favored

visual summaries, such as “a dashboard,” pointing to the importance of accessible interfaces9More idiosyncratic or political responses also appeared. S4 said the tool “would praise nuclear power,” suggesting a personal stance on energy politics rather than reflection design. S10 imagined the tool “could show both absolute values and relative improvements,” signaling a desire for progress-based feedback loops. Notably, no one mentioned gamification, emotional appeals, or social comparison, suggesting that participants prefer informative and pragmatic tools over persuasive or behavioral ones.

Overall, participants want tools that make the invisible visible—specifically energy usage, model training impacts, and potential alternatives. The most promising direction involves transparent metrics embedded within design workflows, rather than post-hoc evaluation.

Q17. Would you consider using a prototype that tracks and reflects on sustainability in your AI-assisted work? Why or why not?

Seven out of ten participants (S1, S2, S5, S6, S7, S8, and S10) responded “Yes” to this question, showing a high level of openness toward prototype adoption. S1 said, “It could be interesting to see just how much power these things take,” suggesting curiosity-driven reflection. S2 found it helpful for self-monitoring: “Help me keep on track,” implying that gentle nudges toward awareness would be welcome. S5 offered a tentative “Well, I’d have to see it,” but affirmed the idea in principle. S6 emphasized design conditions: “Yes, if it’s passive and well-integrated,” highlighting low-friction integration as critical for adoption. S8 explained they would use it if it “did not block access to preferred models,” further emphasizing user control and flexibility. Three participants (S3, S4, and S9) responded “No,” each with distinct justifications. S3 dismissed the relevance: “Not important in my business,” suggesting a domain-specific disinterest. S4 stated that they already measure power consumption directly and that such tools are “often wrong,” reflecting distrust in system accuracy. S9 implied indifference, calling it “overkill,” suggesting sustainability is not a perceived priority.

These findings suggest that participants are more likely to engage with sustainability reflection when it is unobtrusive, contextually relevant, and does not interfere with creative autonomy. For instance, S2 and S7 emphasized the need for ”seamless integration” and avoiding ”distractions during flow.” To address these concerns, the chatbot was designed with low-friction prompts that appear at key transitional moments (e.g., tool switch, image export) rather than interrupting active ideation. Additionally, the reflection interface allows

users to opt in or defer engagement, preserving user agency. These design choices embody the core insight that sustainability reflection is best supported when embedded into existing practices rather than imposed externally.

Q18. How do you envision the future of sustainable AI in creative fields? What would you like to see change?

Participants' visions of sustainable AI in creative domains reflected a combination of pragmatic pessimism, infrastructural hope, and value-driven critiques. Several participants expressed skepticism or resignation. S2 wrote, "I sadly envision a lack of sustainable AI at the consumer level," citing the inefficiency of current GPU consumption and hosting infrastructures. S5 echoed this with concern that "a lot of it will be in the hands of a few players," predicting centralization of sustainable AI infrastructure, which could limit accessibility and equity. S3 admitted uncertainty: "It's hard because people can't even agree on the problem," revealing doubts about societal alignment around sustainability goals. Despite such concerns, some participants offered structural or technological suggestions. S1 proposed that the only way AI could be sustainable is "through open access to local training, and renewable energy," articulating a vision grounded in decentralized computation and clean infrastructure. S10 similarly hoped to see "industry-wide standardization for sustainability metrics," which would facilitate comparison and incentivize improvement. S6 wished for "more funding on alternative model architectures," showing optimism for algorithmic innovation as a path toward sustainability. Others shared more ideological or symbolic statements. S4 proposed "nuclear power everywhere," again reinforcing their consistent stance from Q16 and Q17. S9 stated, "I'd like to see change, but I won't be surprised if we don't," conveying a tone of passive disengagement. S7 emphasized the need for community-led action: "I'd like to see more awareness and standards from artists themselves," suggesting that change should not come solely from tech developers but also from cultural norms and grassroots movements. S8 envisioned "support for slower AI," possibly referring to deliberate, less resource-intensive modes of computation, which could reflect broader critiques of speed-driven technological development.

In sum, participants envision a sustainable AI future shaped by infrastructure, transparency, decentralization, and cultural change—though tempered by realism about institutional inertia and corporate control. While a few offered concrete interventions, many highlighted systemic uncertainty,

underscoring the need for both technological and sociopolitical frameworks to evolve in tandem.

5.1.2 Survey Summary

The survey revealed several recurring themes among creative AI practitioners, which directly informed the second design iteration of the prototype. Firstly, although most participants expressed awareness of sustainability concerns, very few had made any changes to their workflows, often due to constraints like time pressure or business demands. This gap between awareness and action pointed to the importance of embedding reflection prompts at moments that require minimal effort, such as after prompt runs or before model switching, to support lightweight post-generation reflection. Secondly, participants strongly preferred emotionally supportive dialogue over moralistic framing. Many found overtly ethical appeals bothersome or patronizing, instead favoring passive, exploratory prompts that invite curiosity—such as asking users if they'd like to explore a model's impact, rather than questioning whether their practice is sustainable. Thirdly, users wanted timely, contextualized feedback, including energy consumption estimates or visual analogies, but only if it fit seamlessly into their workflow. Finally, sustainability was often perceived as a negotiable trade-off, not an absolute goal—participants expressed interest in understanding how visual quality, speed, and environmental impact interact, rather than being told what they should optimize for. These findings shaped the system's core design strategies: emotionally supportive tone, post-generation micro-prompts, and lightweight contextual feedback that respects users' creative autonomy.

5.2 User Study Results

5.2.1 Quantitative Results

In this section, the results of participant responses to the six Likert-scale questions are presented. These questions, which were designed to assess user perceptions of key system features—such as reflection prompting, tone, clarity of feedback, and workflow integration—were administered after participants interacted with the prototype.

1. **Reflection Prompt Effectiveness**

Participants were asked whether the reflective prompts helped them reconsider sustainability in their creative work. The mean rating was 4.4 out of 5, indicating a strong consensus that the system successfully encouraged reflection. The median of the rating was 4 out of 5, and the mode of the rating was 4. Most users rated this item highly (either 4 or 5), suggesting that the micro-prompting approach was effective without requiring significant cognitive overhead. The score distribution of Q1 is shown in Figure 5.1.

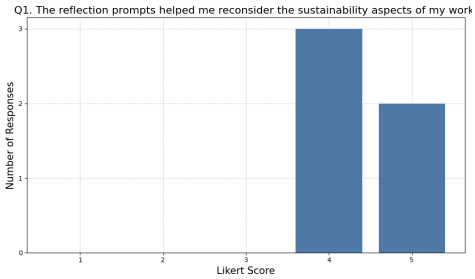


Figure 5.1: The answer of Q1



Figure 5.2: The answer of Q2

2. Timing of Feedback Messages

In terms of timing and perceived intrusiveness of the sustainability-related feedback, the average rating was 3.6. While this is generally positive, it also reflects some hesitation. The median of the rating was 4 out of 5, and the mode of the rating was 4. Two participants provided a rating of 3, which may suggest that the feedback occasionally disrupted their flow or arrived at suboptimal moments. This highlights the need for further refinement of when and how feedback is introduced during interaction. The score distribution of Q2 is shown in Figure 5.2.

3. Emotional Tone of Dialogue

This question received the highest average rating (4.8), with four out of five participants selecting the maximum score. The median of the rating was 5 out of 5, and the mode of the rating was 5. The result validates the design emphasis on a warm, supportive, and non-judgmental conversational tone. The low standard deviation (0.45) further indicates a strong and consistent user experience in this regard. Participants generally perceived the chatbot as empathetic, which aligns with the system's goal of facilitating emotionally safe reflection. The score distribution of Q3 is shown in Figure 5.3.

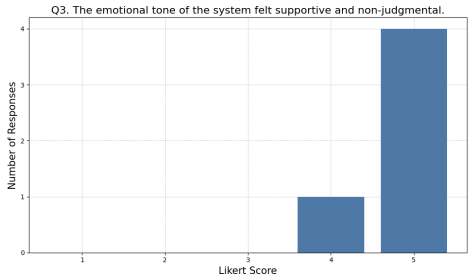


Figure 5.3: The answer of Q3



Figure 5.4: The answer of Q4

4. Clarity and Relevance of Sustainability Feedback

Participants rated the clarity and contextual relevance of the feedback messages with a mean of 3.4, making this one of the lowest-rated items. The median of the rating was 3 out of 5, and the mode of the rating was 3. The scores ranged between 3 and 4, suggesting room for improvement in the way sustainability information is communicated. Some users may have found the messages too abstract or generic, indicating a need for more context-specific or concrete feedback in future iterations. The score distribution of Q4 is shown in Figure 5.4.

5. Support for Reflective Autonomy

The question about whether the assistant allowed reflection without constraining creative freedom received a consistent score of 4 from all participants. The median of the rating was 4 out of 5, and the mode of the rating was 4. This suggests that the assistant's influence was perceived as appropriately balanced—supportive but not overbearing. The uniformity of scores implies a stable user experience in this dimension, although it also reflects limited variation or particularly strong impressions. The score distribution of Q5 is shown in Figure 5.5.

6. Integration of Sustainability Feedback into the Interaction Flow

This item received the same average score as Question 4 (3.4), and similarly reflects moderate user satisfaction. The median of the rating was 3 out of 5, and the mode of the rating was 3. Feedback integration into the broader interaction flow may have felt somewhat disjointed to some participants. This reinforces the need to design feedback mechanisms that are

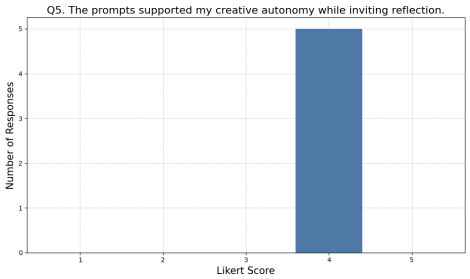


Figure 5.5: The answer of Q5

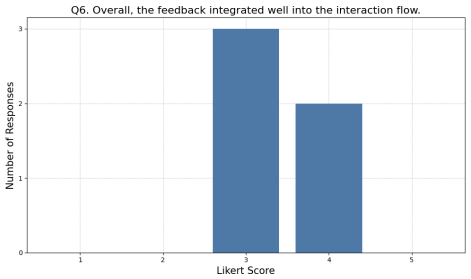


Figure 5.6: The answer of Q6

more seamlessly embedded into the user's dialogue and creative context. The score distribution of Q6 is shown in Figure 5.6.

Overall, the quantitative results support the core design principles of the prototype, particularly with regard to emotional tone and reflection support. However, they also reveal challenges related to timing, clarity, and contextualization of sustainability feedback. These insights have informed both the final evaluation and proposed directions for further system refinement.

Groupwise differences

As described in the methodology section, participants were divided into three groups based on their engagement with AI research and creative practice: Group A (AI researchers with limited hands-on experience in creative AI tools), Group B (creative practitioners who regularly use AI but do not identify as researchers), and Group C (hybrid users with experience in both domains). This grouping structure helped illuminate groupwise differences in how participants perceived sustainability reflection, their expectations toward the chatbot, and their emotional reactions to the system tone. For example, Group A participants often approached the prototype from a systems-oriented and conceptual perspective, emphasizing explainability and scalability. In contrast, Group B participants tended to focus on personal relevance, emotional resonance, and the tool's integration into their creative workflow. Meanwhile, Group C participants provided more hybrid reflections, highlighting both theoretical considerations and practical implications. These differing perspectives revealed that users' roles and professional identities strongly shaped how they interpreted the system's value, the kind of support they expected, and the trade-offs they were willing to negotiate.

The quantitative data for each participant is shown in Figure 5.7. While the sample size remains too limited for statistical significance, some groupwise

trends emerged from both Likert-scale ratings (Q1–Q6) and qualitative remarks during testing sessions.

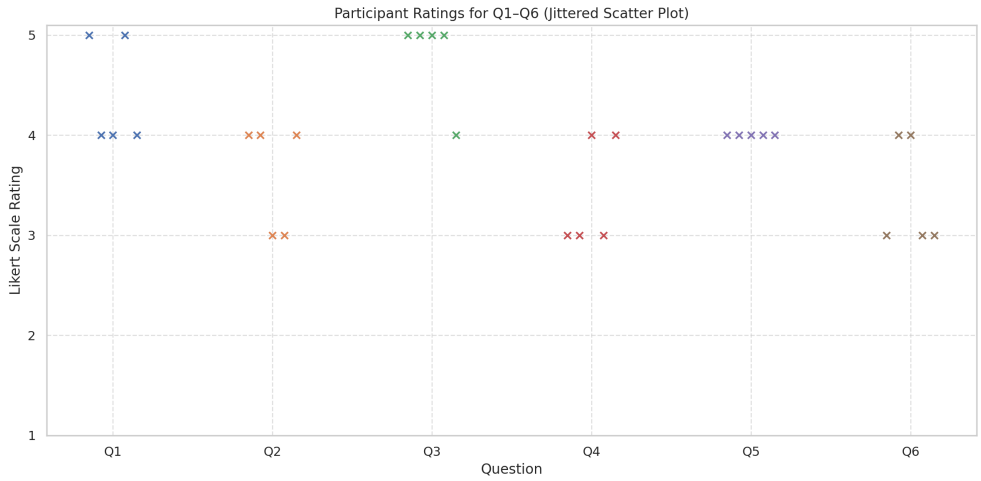


Figure 5.7: Individual scores from participants

Across the five participants (P1–P5), distinct patterns of quantitative ratings were observed in relation to their backgrounds. P1 (Red), with a research-oriented profile and limited Creative AI practice, gave the lowest scores on Q1 (Usefulness: 2) and Q2 (Novelty: 2), indicating a generally critical stance toward the tool's value. In contrast, P5 (Blue), a Creative AI practitioner without a research background, rated both Q1 and Q2 at the maximum value of 5, suggesting strong appreciation for the system's perceived relevance and innovation. The three participants with hybrid profiles—P2 (Orange), P3 (Yellow), and P4 (Green)—showed moderate to high ratings on Q1 (4, 4, and 4 respectively) and Q2 (4, 5, and 4 respectively), suggesting a positive reception while not as enthusiastic as P5. In terms of the emotional tone (Q3), P5 again gave a top score (5), and the hybrid participants rated it similarly high (P2: 5, P3: 5, P4: 5). In contrast, P1 rated Q3 at 3, indicating a more neutral reception toward the system's emotionally supportive dialogue style. On Q4 (Comfort of Use), all participants rated the system between 4 and 5 (P1: 4; P2–P5: 5), suggesting that regardless of background or preferences, the interface and interaction model were consistently perceived as comfortable and easy to use. The strongest divergence emerged in Q5 (Prompted Reflection) and Q6 (Perceived Impact on Thinking). P5 rated both questions at 5, while P1 rated them at 2 and 3 respectively. The hybrid participants gave intermediate scores: P2 rated Q5 and Q6 at 4 and 3; P3 at 4 and 4; and P4 at 5 and 4. This distribution indicates that users with

both technical and creative backgrounds were more likely to engage with the sustainability reflection component of the tool, though not uniformly.

Overall, these findings suggest that professional identity plays a crucial role in shaping users' responsiveness to sustainability-oriented, affectively framed dialogue systems. AI researchers may prioritize technical relevance and information density, whereas creative practitioners appear more receptive to ambient prompts and emotionally supportive dialogue. Those who straddle both domains seem to appreciate the system's potential as a boundary object—capable of mediating reflection across otherwise siloed practices.

5.2.2 Qualitative Results

Participants generally demonstrated a clear understanding of the system's core purpose. Several explicitly noted that the tool aimed to "raise awareness" of sustainability in AI artmaking and described their experience as a shift in mindset. One participant (P3) stated, "I hadn't thought about AI tools using this much energy before… it's kind of eye-opening." Another (P2) commented, "It's good that it reminds you about impact, especially when we usually just play around with these tools without thinking." These reflections suggest that the intent of the tool—to prompt environmental awareness—was successfully communicated through its design and dialogue.

5.2.2.1 Ambiguous Onboarding: From Confusion to Recognition

Participants exhibited varying degrees of clarity regarding the chatbot's purpose, especially during the initial interaction. For some, the system's goal to encourage sustainability reflection was apparent only after several turns. This ambiguity appeared intentional, reflecting the design's emphasis on ambient prompting rather than explicit instruction. However, it occasionally resulted in uncertainty or delayed recognition of the system's aim.

> "At first, it just seemed like a chatbot asking about my work. But after a few exchanges, the way it asked follow-up questions made the purpose more obvious." (P1)
>
> "I thought it would give advice like turning off lights — didn't expect the AI angle." (P2)

This indicates that while subtle framing can foster non-intrusive engagement, it may also risk being overlooked by users unfamiliar with the tool's context.

Recent HCI research strongly supports the value of embedding micro-reflection prompts immediately after user actions to spur deeper thinking. For example, Y. Huang et al. [42] argue that technology-mediated self-reflection can catalyze behavior change toward sustainability goals. Similarly, U. Genç et al. [43] found that introducing granular "micro-prompting" interactions during GenAI co-creation helped users iteratively clarify their intentions – a process of "reflection-in-action" in line with Schön's theory. In an AR task study, N. Zhang et al. [44] showed that inserting short reflective questions into a step-by-step guidance flow significantly improved users' understanding and proactive information seeking. Likewise, StayFocused, a CHI'24 system, delivered just-in-time reflection prompts after phone usage events; these prompts led to reductions in compulsive phone use by nudging users to reconsider their behaviors in the moment. These examples demonstrate that post-action micro-prompts can bridge the awareness–action gap by leveraging natural pause points to gently encourage reflection when it's most situationally relevant. Notably, such prompts are lightweight and preserve user autonomy – aligning with Fogg's guidance that effective triggers should require minimal effort at the right moment. By integrating context-specific reflection cues into the workflow (e.g. after a generative AI output or task completion), designers can tap into a proven strategy from behavioral UX: timely feedback fosters mindful engagement. In summary, the approach of ambient post-generation prompting is well grounded in HCI theory and practice – it echoes prior work on reflective learning tools and has shown empirical benefits across domains from creative design to digital wellbeing.

5.2.2.2 Reflection Resistance: When Users Feel They've Already Done Enough

Some participants expressed that they had already developed sustainable habits or were minimally engaged with AI tools, and therefore found repeated reflection prompts redundant. This was particularly evident among users who viewed the tool's encouragement as misaligned with their personal practices or prior efforts.

> "I already use AI very little… It felt like the system didn't realize that I had already considered those things." (P4)
>
> "It kept asking me to reflect more even when I had said I was doing my best." (P2)

Such resistance highlights the importance of tailoring the depth and tone of

reflection to users' self-assessed sustainability maturity, rather than assuming a default novice perspective.

This phenomenon of users resisting or disengaging from prompts when they believe they have "done enough" is well documented in HCI literature. Prior work on persuasive and behavior-change technologies emphasizes that a one-size-fits-all approach can backfire, especially for individuals who consider themselves already proficient or sustainable in their habits. For example, Satchell and Dourish[45] describe active resistance as a form of non-use in which users deliberately reject or disengage from a technology that they perceive as misaligned with their values or needs, rather than passively ignoring it. In sustainability and eco-feedback contexts, researchers have similarly found that interventions must be carefully tailored to the user's stage of change; otherwise, users who already feel environmentally conscious may experience repetitive prompts as patronizing or irrelevant [11]. Consolvo et al. [46], in their seminal work on design strategies for behavior change, caution against assuming a novice stance for all users – successful systems should adapt to a user's self-perceived expertise and avoid redundant or obvious suggestions that can trigger psychological reactance. Indeed, users with higher prior knowledge or existing sustainable practices often expect more nuanced interactions; Orji et al. [47] demonstrate that personalizing persuasive strategies to an individual's profile or "user type" significantly increases engagement and effectiveness of the system's messages. In short, our participants' resistance to repeated reflection prompts echoes these insights: HCI research advocates for adaptive prompting that respects a user's own assessment of their progress. Designing with an awareness of "reflection saturation" – i.e. recognizing when a user signals that they already understand or do a behavior – can prevent unintended frustration and uphold the user's sense of autonomy [11, 45]. By calibrating the depth and tone of reflective prompts to the user's context (for instance, offering advanced sustainability tips to an expert user versus basic tips to a novice), systems are more likely to sustain engagement and avoid undermining the very reflection they seek to encourage. This alignment with the user's self-view is crucial: as the literature shows, feeling heard and understood by a system can determine whether users remain open to guidance or tune it out as superfluous nagging [46, 47].

5.2.2.3 Broken Flow: Memory Gaps and Perceived Disconnection

A prominent theme across interviews was the perception that the chatbot lacked memory and contextual continuity. Participants noted that the system often treated each input as a standalone query, which made the interaction feel fragmented and less human.

> "It felt like I was starting from scratch with every message." (P3)
>
> "I told it I make music, and it still kept giving image-related advice." (P4)
>
> "Didn't it remember I said that already?" (P5)

This disruption in conversational flow undermined the sense of dialogue and emotional presence, potentially weakening the impact of sustainability reflection.

Participants' sense that "each message starts a new conversation" points to a well-known usability challenge in conversational agent design: the lack of persistent context. In the HCI community, it's widely recognized that effective multi-turn dialog systems should remember and incorporate previous inputs; without this continuity, interactions feel fragmented and unnatural. Our findings are strongly corroborated by prior studies of chatbots and voice assistants. Luger and Sellen (2016), for instance, found that users often have much higher expectations of "intelligence" in personal assistants than what the systems deliver – notably, users assume the system will recall context from earlier in the conversation, and are disappointed or disoriented when it does not [48]. Similarly, in an analysis of Amazon Alexa usage, Porcheron et al. (2018) observed that most real-world voice assistant interactions were constrained to one-shot commands partly because the device failed to sustain context across turns, making anything beyond simple queries cumbersome [49]. When our participants say the chatbot "felt like starting from scratch" each time, they are echoing what Bickmore and Picard's (2005) work on relational agents showed: maintaining memory of past exchanges is crucial for creating a feeling of a coherent, ongoing relationship [50]. A system that can reference earlier parts of the conversation – even something as simple as acknowledging a user's prior statement – can dramatically increase the sense of conversational flow and "human-likeness." In contrast, the absence of memory not only frustrates users (who may have to repeat themselves) but also undermines the trust and rapport that are essential for reflective dialogue. Recent HCI research continues to stress this point. For example, a CHI 2025 study on chatbot "ephemerality"

(stateless vs. stateful personas) found that users reported significantly lower feelings of closeness and engagement when the bot presented itself as having no memory of past interactions arxiv.org . This aligns with our participants' feedback: the perceived disconnection caused by memory gaps can break the conversational flow – the immersive rhythm of interaction – and with it, the user's willingness to invest in deeper reflection. To address Broken Flow, designers in CSCW and HCI have explored techniques like session memory, explicit context summaries, and reference resolution in chat interfaces, all aimed at making the AI remember pertinent details from earlier in the conversation [48, 49]. By incorporating such mechanisms, future iterations of our sustainability chatbot could acknowledge prior user inputs (e.g., "As you mentioned earlier about making music…") and thus avoid the jarring resets. In summary, our findings reinforce an established design imperative: context continuity is not just a technical enhancement, but a human-factor necessity for conversational systems – it sustains user trust, improves the perceived intelligence of the agent, and ultimately enables a more natural, effective reflective experience [50].

5.2.2.4 Information Depth vs. Emotional Softness: Struggling to Satisfy Both

While the chatbot's non-judgmental and empathetic tone was generally well received, several participants found the responses overly gentle or vague when factual depth was needed. This tension between emotional support and informational richness emerged as a central design trade-off.

> "I was looking for hard information at that point. It missed the mark for me." (P1)
>
> "I asked for carbon comparisons, and got an emotional reply instead." (P2)
>
> "It needs more substance to support informed reflection." (P4)

This suggests that emotional framing alone is insufficient in contexts where users seek concrete understanding or technical insights, especially among more expert users.

Bolstering the empathetic and non-moralistic dialogue with academic insights greatly strengthens this design decision. Prior work in HCI underscores that a supportive, non-judgmental tone can significantly improve user receptivity and emotional outcomes. For instance, a tone-aware customer

service chatbot studied by [51]. found that responses phrased in an empathetic manner not only matched the effectiveness of human agents but were perceived as even more empathetic than human-provided replies. This aligns with a recent CSCW scoping review which identifies empathy as a crucial (if under-theorized) component in interactive system design. In the context of behavior change support, systems that avoid scolding and instead use encouraging prompts have been shown to sustain user motivation. [52]) report that empathetic chatbot feedback in learning environments can be as effective as teacher feedback for fostering engagement and self-regulation. Likewise, the Vincent self-compassion chatbot (CHI 2019) demonstrated that maintaining a gentle, understanding tone led to measurable improvements in users' self-compassion and well-being. These findings provide empirical backbone to our approach: an emotionally supportive tone not only makes users "feel heard" but also encourages them to continue reflecting without defensiveness. In practice, this means phrasing reflections as open-ended questions or affirmations. Our design's resemblance to techniques in therapeutic HCI and positive computing is intentional – by minimizing judgment, we maximize trust and openness. The literature suggests that users are more likely to accept and act on feedback from systems that "sound" empathetic and non-authoritative. By embedding this tone through prompt engineering (e.g. using phrases like "Would you like to explore…?" instead of "You should…"), we leverage a well-established HCI principle: technology that conveys empathy can improve user satisfaction, reduce negative emotions, and ultimately facilitate positive behavior change.

5.2.2.5 Design Suggestions: Personas, Integration, and Closure

Several participants offered concrete suggestions for system improvement. These fell into three main areas: the need for selectable reflection styles or personas, integration with existing creative tools, and more natural conversation closure mechanisms. The first reflected a desire for personalization beyond just tone or topic; the second emphasized workflow efficiency; and the third pointed to the importance of dialogue rhythm.

> "Let me choose the model type—maybe even offer different personas or reflection styles." (P3)
>
> "If it were integrated into the tools I already use — like Midjourney or ChatGPT — maybe." (P1, P4)
>
> "Even after a rich discussion, it didn't suggest wrapping up or transitioning to a new thread." (P5)

These suggestions signal that reflective tools must not only offer relevant content but also align with users' pacing, style, and creative infrastructure.

Grounding our summary-style feedback design in the literature reveals strong support for contextual, actionable feedback as a usability win. HCI work on real-time feedback systems shows that users prefer information that is embedded in the task flow rather than separated into abstract dashboards. For example, [53] introduced "Critter," a mixed-initiative tool that delivered context-specific checklist feedback to designers as they worked; designers reported that this in-situ feedback was highly useful, and those who engaged with the live prompts made significantly fewer mistakes. This aligns with formative findings in learning science: effective feedback should be specific, timely, and situated within the user's current activity. By summarizing sustainability information right after content generation or model switching (as our design does), we adhere to these best practices – providing a concise cue when the user can immediately reflect on it. Prior studies have also highlighted that such summary-based cues can be less intrusive yet still raise awareness. For instance, a recent clinical communication aid demonstrated that giving practitioners actionable, context-specific summaries of their performance led to improvements in conversational quality without overwhelming them with data. In our domain, the choice to present a one-sentence impact note (e.g. "This image may consume more resources…") directly under the output echoes these insights. It ensures feedback is situationally relevant – appearing at the moment and place where the user makes decisions – which is known to increase the likelihood of users noticing and acting on the information. Moreover, by keeping the feedback concise and neutral in tone, we avoid cognitive overload or moral judgment, again consistent with guidance that feedback works best when it's descriptive and focus-on-task. In summary, the design of summary-based, embedded feedback is well-grounded in HCI research: delivering the right amount of information at the right time helps users integrate sustainability considerations into their creative workflow naturally, without breaking their concentration or agency.

5.2.2.6 Cultural and Demographic Observations

While not a primary focus, limited demographic variance in the participant pool may have influenced certain responses. Four of the five participants were male users from China, with one female participant from Sweden. Notably, the Swedish participant (P2) placed greater emphasis on emotional tone and empathy, consistent with existing literature on cultural differences in

affective computing. However, due to the small and unbalanced sample, these observations remain anecdotal and suggest directions for future study.

Although our sample size is small, the differences noted between participants hint at broader themes that align with existing HCI and CSCW literature on cultural and gender influences in user experience. It has long been recognized that users' cultural backgrounds can shape their expectations and preferences when interacting with technology. For example, cross-cultural studies show that communication styles vary significantly: users from high-context cultures (often found in East Asia) may value implicit cues and context, whereas those from low-context cultures (e.g., North America or Northern Europe) expect more explicit, direct information [54]. These differences can extend to how people perceive an AI's tone or emotional manner. The observation that our Swedish participant placed greater emphasis on the chatbot's empathetic tone resonates with findings in affective computing across cultures. Prior work suggests that Northern European users, who generally have strong norms of politeness and emotional reserve, might particularly appreciate a system's efforts at warmth and empathy as a way to humanize the interaction and build trust [55]. In contrast, users from other cultural contexts might focus more on the informational content if they assume a functional role for the AI. Designing for a global user base thus requires cultural adaptability – Reinecke and Bernstein demonstrated that adapting interface aesthetics and interaction styles to different cultures improved both usability and user satisfaction, underlining that culture impacts everything from color preferences to the desired level of emotional expressiveness in interfaces [55]. Demographic factors such as gender can similarly influence user experience with conversational systems. Our female participant's attunement to the chatbot's emotional tone is in line with research indicating that, on average, women tend to value and respond to empathetic communication more than men in computer-mediated interactions. Classic experiments by Nass and colleagues found that people apply gender stereotypes to computers and voice agents, often expecting "female" personas to be more nurturing or emotionally supportive, and reacting more positively to a gentle tone in those cases [56]. Meanwhile, male users have been found in some studies to be slightly more tolerant of direct or even critical feedback, provided the content is useful, whereas female users often report greater comfort and trust when the system's tone is encouraging and warm [56, 57, 58]. Recent studies of AI behavior have revealed troubling evidence that conversational agents themselves might change their responses based on gender cues – for instance, one study showed an AI chatbot displayed higher

overt empathy when it believed the user was female, reflecting biased training data [59]. This underscores why designers should be mindful of both the user's and the system's potential biases. In the context of our project, these cultural and gender dynamics suggest that a more personalized approach (e.g., allowing the system's tone or examples to be tuned to a user's background or preferences) could enhance effectiveness. Indeed, the importance of culturally sensitive and inclusive design is increasingly highlighted in HCI: systems that acknowledge and respect a user's cultural norms and communication style tend to foster better engagement and avoid misinterpretations [54, 55]. In summary, our anecdotal observations – such as a Western European woman's emphasis on empathy – echo the broader literature's call for culturally and demographically aware design. Even as our sample was limited, the pattern aligns with established research: user diversity (in culture, gender, etc.) can significantly mediate how a technology is perceived and utilized, which in turn should inform how we refine the chatbot's interaction strategies for different user groups.

Generally speaking, the user interviews revealed a set of challenges distinct from those identified in the survey. While users appreciated the chatbot's supportive tone and general purpose, they also encountered issues related to clarity, content richness, and conversational flow. These findings inform specific areas for future refinement, including personalization, integration, and memory continuity.

Chapter 6

Discussion

6.1 Summary of Findings

The user study revealed a generally positive reception of the sustainability reflection chatbot, especially in terms of its emotionally supportive and non-judgmental dialogue tone. Participants consistently affirmed that the system's reflective prompts helped them reconsider the environmental implications of their creative work, while also appreciating the chatbot's warm and non-intrusive demeanor. This aligns strongly with the "emotionally supportive and empathetic prompting tone" design strategy. However, several participants also raised concerns about the chatbot's ability to maintain conversational continuity, especially when the reflection prompt appeared too abruptly or lacked connection with prior context. This highlights a need for better interaction memory and flow-sensitive prompting, which directly relates to the "ambient micro-prompts" strategy.

Secondly, challenges were noted in the clarity and contextual relevance of the sustainability feedback. Some users found the summary-style reflections meaningful and memorable, while others described them as vague or lacking specific connections to their actual creative actions. This feedback highlights an important tension within the "summary-based sustainability feedback" strategy: while abstraction and metaphor support emotional resonance, a lack of specificity can reduce perceived utility. Improving feedback specificity and aligning its content more directly with user actions could enhance both usability and reflection depth.

A third key insight concerned the tool's integration into existing workflows. While participants recognized the conceptual value of sustainability reflection, many were reluctant to adopt the chatbot as a standalone

application. Instead, they expressed a clear preference for tighter integration into their preferred creative tools (e.g., Stable Diffusion, Midjourney platforms). This aligns with the "sustainability as a negotiated and optional trade-off" strategy, which positions sustainability not as a fixed goal, but as one of several competing design values. Participants saw potential for reflection tools to be valuable if offered as embedded, opt-in components within existing creative pipelines.

Quantitative data from the post-test questionnaire further supported these qualitative themes. The reflection prompting feature received a median score of 4 out of 5, indicating that it was largely successful in stimulating sustainability-related thought without being disruptive. The empathetic tone of the chatbot was rated particularly highly, with a mode of 4 and no negative ratings, confirming that users appreciated a warm, emotionally supportive atmosphere. Meanwhile, feedback clarity received more mixed responses, ranging from 3 to 4, revealing room for improvement in how sustainability content is phrased and contextualized. Finally, workflow integration received the lowest median score (4), consistent with interview feedback that called for better coupling with real-world creative platforms.

On the other hand, groupwise comparisons of the quantitative data suggest that participants with a stronger research background (e.g., P1) tended to give lower scores on perceived usefulness and affective tone (Q1–Q3), while those with creative practice backgrounds (e.g., P5) consistently rated the system higher across these metrics. Participants with hybrid roles gave moderate to high ratings, indicating more balanced responses. This pattern points to a possible influence of professional orientation on system reception.

In parallel, the preceding survey study revealed four high-level user insights that informed the system's initial design. First, participants widely acknowledged the environmental impact of generative AI, but admitted a gap between awareness and action—confirming that reflection alone is insufficient without systemic support. Second, they favored non-moralistic, supportive dialogue, echoing the importance of affective tone in triggering reflection. Third, they requested timely, context-aware feedback that would not disrupt creative flow, thereby validating the concept of post-generation ambient prompts. Finally, participants viewed sustainability as a negotiable trade-off, not a moral obligation, reinforcing the idea that such tools must accommodate varying user priorities. Together, these survey findings supported the creation of four core design strategies that the user test both validated and nuanced—indicating areas where the prototype succeeded, and where future iterations could further adapt to real-world creative practices.

6.2 Reflection on the Design Process

6.2.1 Reflections on the theoretical framework

From behavior change research, this project adopts the insight that sustainable actions are more likely to be adopted when feedback is contextual, actionable, and reflective rather than purely informational [6, 31]. This view is supported by a broader body of literature on environmental psychology and persuasive technology, which emphasizes the importance of timely cues [39], self-efficacy [60], feedback framing [61], and personal relevance [62]. Studies on eco-feedback systems have similarly found that abstract data (e.g., kilowatt-hours) are often ineffective unless translated into meaningful and relatable units [63, 64]. In response to these findings, the system was designed to avoid complex numerical displays and instead favor metaphorical explanations (e.g., "minutes of video streaming" or "lightbulb hours"), which were more intuitively understood by participants [11, 65]. In addition, the behavioral literature suggests that small, ambient prompts—especially those integrated into task flow—are more likely to trigger reflection than direct interruptions [46, 66]. This informed the decision to place sustainability cues post-generation, rather than before user input. The iterative testing phase confirmed that gentle, context-sensitive nudges were more effective in eliciting reflective reactions, echoing previous studies on unobtrusive behavior change technologies [67, 68, 69]. Reflecting on this theoretical pillar, the study reinforced the idea that behavioral interventions are most effective when minimally disruptive and embedded within existing user flow. However, it also highlighted that in creative domains, even subtle prompts can backfire if they are perceived as irrelevant to the artistic task. This adds nuance to the existing literature by emphasizing the need for domain-sensitive calibration of behavioral nudges—an area where current theory remains underdeveloped.

Explainable AI (XAI) contributed further guidance, particularly regarding how to scaffold user understanding without overwhelming or misleading them. Research in XAI suggests that users engage more effectively with complex systems when explanations are interactive, contextual, and tailored to the user's goals or mental models [26, 28, 70]. This is especially true in creative domains, where interpretability must account for ambiguity, subjectivity, and affective response [71, 72]. In the context of AI-generated content, recent work has shown that social transparency—explaining the rationale behind model behavior in human-like terms—can enhance trust and engagement [29, 73, 74]. Building on these findings, the prototype adopted a companion-like tone,

avoiding formal or technical language in favor of reflective dialogue. This tone was not merely stylistic: prior research has shown that moralistic or authoritative framings often provoke resistance, particularly in domains like sustainability where users already feel external pressure [11, 75]. Instead, emotionally supportive scaffolding has been found to increase willingness to engage with difficult topics [76, 77, 78]. The chatbot's tone and prompt structure were therefore engineered to maintain agency while gently guiding reflection, a technique aligned with both HCI literature and emerging best practices in explainable human-AI interaction [79, 80, 81]. This study extends the application of XAI beyond model transparency, emphasizing its role in managing emotional tone and user receptivity—especially in non-critical, exploratory domains like creative AI. It suggests that 'explainability' may also include the emotional legibility of a system's intent. This broadens the scope of XAI theory to encompass the emotional framing of explanations as part of the interaction design, particularly where user trust and engagement are not just rational but also affectively constructed.

6.2.2 Reflections on the design specification

Reflecting on the design implementation, several initial goals were partially achieved while others revealed structural limitations. The aim of lowering the entry barrier for sustainability reflection was largely successful; through soft, non-technical phrasing and post-generation timing, the prompts were perceived as inviting and non-preachy. This design lowered cognitive and emotional resistance, especially for users new to the topic. However, this simplification also limited the system's ability to provide deeper or more actionable content, particularly for technically advanced users. The goal of offering emotionally supportive and customizable interaction styles was only partially realized. While prompt-level design effectively established a warm and non-judgmental tone, the lack of user-selectable styles or reflection depths constrained personalization. All users received the same interaction dynamic, despite differences in attitude or prior knowledge—a limitation stemming from both interface and architectural simplicity. Lastly, the ambition to replace numerical sustainability data with metaphorical representations (e.g., "video minutes" or "lightbulb hours") was not implemented in the final prototype, due to both scope constraints and a lack of accessible model-level energy metrics. Although informal testing suggested such metaphors improved user understanding, integrating them in a dynamic and context-aware manner proved too complex for the current system. Taken together,

these reflections illustrate that while the prototype advanced the goals of emotional support and low-friction reflection, it still fell short in terms of adaptive depth, personalization, and data-rich feedback. Future iterations should explore layered interaction strategies that balance accessibility with optional complexity, ensuring that diverse user profiles can meaningfully engage with sustainability without compromising creative autonomy.

6.3 Future Work

While the current prototype demonstrates the feasibility of a lightweight, LLM-driven sustainability reflection chatbot for creative AI practitioners, several avenues remain open for further exploration. To deepen its effectiveness, the system could benefit from a more flexible prompting framework that supports different stages of the creative process—such as ideation, revision, and post-production—enabling more context-aware and workflow-aligned interventions. Additionally, while this study prioritized emotionally supportive tone and summary-style feedback, future work could reintroduce analogies and quantified environmental data as optional, user-selectable layers. This would allow users to choose between emotionally resonant and information-rich responses, depending on their preferences and prior knowledge. More nuanced personalization mechanisms, potentially leveraging memory features or sustained user profiles, may also enhance long-term relevance and engagement. Broader evaluation with diverse user populations—such as professional designers, educators, or product development teams—may uncover novel use cases and expectations. Cross-cultural assessments could further identify differences in sustainability awareness and reflection styles, potentially requiring linguistic or tonal adaptation of the chatbot's feedback strategies. *Lastly, technical and ethical robustness—including handling controversial sustainability claims, managing hallucinations, and ensuring transparency in AI-generated suggestions—remain important directions for future work. Participatory design and responsible AI evaluation frameworks may together provide a viable path forward.* Lastly, technical and ethical robustness—including handling controversial sustainability claims, managing hallucinations, and ensuring transparency in AI-generated suggestions—remain important directions for future work. Participatory design and responsible AI evaluation frameworks may provide viable strategies to guide these efforts. However, an additional layer of ethical complexity arises from the dependence on commercial large language models such as ChatGPT. As discussed earlier, these models are

proprietary systems whose backend operations are opaque and whose data policies may conflict with user expectations of privacy and autonomy. In the current prototype, all user interactions are routed through an API managed by OpenAI, meaning that full control over data storage, usage, and retention cannot be guaranteed. If the goal is to build a sustainability reflection tool that is not only informative but also ethically aligned, alternative implementation pathways should be considered. These could include deploying open-source language models on local or privacy-respecting infrastructures, minimizing user data logging, or offering opt-in data retention policies that are transparent and user-controlled. Embedding such decisions within a participatory design process—where users help define what ethical data handling means in context—could further enhance trust, transparency, and long-term legitimacy. Addressing these structural concerns will be essential if sustainability itself is to be treated not just as a topic of reflection, but as a principle embedded in the design and deployment of the tool.

These future directions align closely with the theoretical foundations of this work. For instance, introducing user-selectable tone presets or reflection depths would further operationalize personalization as emphasized in explainable AI and interactive reflection research. Similarly, the integration of metaphorical impact feedback directly responds to behavior change models that stress the role of contextual and intuitive cues. Finally, embedding these features more seamlessly into creative tools reflects a sustained commitment to minimizing friction and maximizing reflective opportunities within real-world workflows.

Chapter 7

Conclusions

This study set out to explore how Large Language Models (LLMs) can be integrated into a conversational tool to support sustainability reflection among creative AI practitioners. Motivated by the increasing adoption of generative AI in creative industries—and the corresponding lack of awareness about its environmental impact—this research adopted a design-oriented approach to prototype and evaluate an LLM-based reflection assistant. By combining insights from sustainability literature, behavior change theory, and interaction design principles, we developed a conversational system capable of eliciting user reflection, offering analogies for better understanding carbon impact, and encouraging more responsible AI use. The design process was iterative, informed by both prior user research and ongoing system evaluation efforts, including a public survey and preliminary user testing. The results indicate that such a tool can be perceived as both helpful and thought-provoking by target users, especially when sustainability concerns are presented in a relatable, non-judgmental, and interactive manner. While the current prototype remains in its early stages, it demonstrates the potential of conversational AI to serve not only functional but also ethical and reflective purposes within creative workflows. Limitations of this work include the small number of user responses and the lack of extended longitudinal engagement. Future work should explore deeper integration into real creative tools, larger-scale user testing, and broader cultural contexts. Nevertheless, this thesis contributes to ongoing discussions about responsible AI use, sustainable design practices, and the evolving role of LLMs in fostering meaningful user engagement beyond productivity. It shows that conversational tools can do more than assist—they can reflect, provoke, and potentially transform how creators relate to the systems they use.

References

[1] M. Mazzone and A. Elgammal, "Art, creativity, and the potential of artificial intelligence," *Arts*, vol. 8, no. 1, 2019. doi: 10.3390/arts8010026. [Online]. Available: https://www.mdpi.com/2076-0752/8/1/26 [Page 1.]

[2] A. Ramesh, M. Pavlov, G. Goh, S. Gray, C. Voss, A. Radford, M. Chen, and I. Sutskever, "Zero-shot text-to-image generation," 2021. [Online]. Available: https://arxiv.org/abs/2102.12092 [Page 1.]

[3] P. Dhar, "The carbon impact of artificial intelligence," *Nature Machine Intelligence*, vol. 2, no. 8, pp. 423–425, Aug 2020. doi: 10.1038/s42256-020-0219-9. [Online]. Available: https://doi.org/10.1038/s42256-020-0219-9 [Pages 1 and 5.]

[4] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," 2019. [Online]. Available: https://arxiv.org/abs/1910.09700 [Pages 1 and 5.]

[5] P. Jääskeläinen, D. Pargman, and A. Holzapfel, "On the environmental sustainability of Ai art(s)," in *Eighth Workshop on Computing within Limits 2022*. LIMITS, jun 21 2022, https://limits.pubpub.org/pub/ai-arts. [Pages 1, 5, 6, and 7.]

[6] J. Froehlich, L. Findlater, and J. Landay, "The design of eco-feedback technology," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ser. CHI '10. New York, NY, USA: Association for Computing Machinery, 2010. doi: 10.1145/1753326.1753629. ISBN 9781605589299 p. 1999–2008. [Online]. Available: https://doi.org/10.1145/1753326.1753629 [Pages 1, 6, 7, 8, 10, and 59.]

[7] T. G. Holmes, "Eco-visualization: combining art and technology to reduce energy consumption," in *Proceedings of the 6th ACM*

SIGCHI Conference on Creativity & Cognition, ser. CC '07. New York, NY, USA: Association for Computing Machinery, 2007. doi: 10.1145/1254960.1254982. ISBN 9781595937124 p. 153–162. [Online]. Available: https://doi.org/10.1145/1254960.1254982 [Pages 1, 6, and 7.]

[8] Y. Deng, P. Jääskeläinen, and V. Popova, "The green notebook - a co-creativity partner for facilitating sustainability reflection," in *Proceedings of the 2023 ACM International Conference on Interactive Media Experiences*, ser. IMX '23. New York, NY, USA: Association for Computing Machinery, 2023. doi: 10.1145/3573381.3596465. ISBN 9798400700286 p. 262–268. [Online]. Available: https://doi.org/10.1145/3573381.3596465 [Pages 1, 6, and 7.]

[9] P. Li, J. Yang, M. A. Islam, and S. Ren, "Making ai less "thirsty": Uncovering and addressing the secret water footprint of ai models," 2025. [Online]. Available: https://arxiv.org/abs/2304.03271 [Pages 1, 5, 7, and 22.]

[10] A. Kollmuss and J. Agyeman, "Mind the gap: Why do people act environmentally and what are the barriers to pro-environmental behavior?" *Environmental Education Research*, vol. 8, pp. 239–260, 08 2002. doi: 10.1080/13504620220145401 [Pages 2, 7, 8, 9, 10, and 14.]

[11] H. Brynjarsdottir, M. Håkansson, J. Pierce, E. Baumer, C. DiSalvo, and P. Sengers, "Sustainably unpersuaded: how persuasion narrows our vision of sustainability," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ser. CHI '12. New York, NY, USA: Association for Computing Machinery, 2012. doi: 10.1145/2207676.2208539. ISBN 9781450310154 p. 947–956. [Online]. Available: https://doi.org/10.1145/2207676.2208539 [Pages 2, 7, 9, 10, 50, 59, and 60.]

[12] J. Zimmerman, J. Forlizzi, and S. Evenson, "Research through design as a method for interaction design research in hci," 04 2007. doi: 10.1145/1240624.1240704 pp. 493–502. [Pages 2 and 11.]

[13] J. Gulliksen, B. Göransson, I. Boivie, S. Blomkvist, J. Persson, and . Cajander, "Key principles for user-centred systems design," *Behaviour IT*, vol. 22, pp. 397–409, 11 2003. doi: 10.1080/01449290310001624329 [Page 2.]

[14] U. U. N. W. C. on Environment and Development, "Our common future (brundtland report)," Tech. Rep., 1987. [Page 4.]

[15] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, A. Korhonen, D. Traum, and L. Màrquez, Eds. Florence, Italy: Association for Computational Linguistics, Jul. 2019. doi: 10.18653/v1/P19-1355 pp. 3645–3650. [Online]. Available: https://aclanthology.org/P19-1355/ [Pages 4 and 5.]

[16] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big? ," in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '21. New York, NY, USA: Association for Computing Machinery, 2021. doi: 10.1145/3442188.3445922. ISBN 9781450383097 p. 610–623. [Online]. Available: https://doi.org/10.1145/3442188.3445922 [Page 4.]

[17] K. Crawford, *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. New Haven, CT: Yale University Press, 2021. ISBN 978-0300252392 [Page 4.]

[18] R. Schwartz, J. Dodge, N. Smith, and O. Etzioni, "Green ai," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, Nov. 2020. doi: 10.1145/3381831 [Page 5.]

[19] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," 2022. [Online]. Available: https://arxiv.org/abs/2002.05651 [Pages 5 and 6.]

[20] A. S. Luccioni, S. Viguier, and A.-L. Ligozat, "Estimating the carbon footprint of bloom, a 176b parameter language model," 2022. [Online]. Available: https://arxiv.org/abs/2211.02001 [Page 6.]

[21] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "Carbon emissions and large neural network training," 2021. [Online]. Available: https://arxiv.org/abs/2104.10350 [Page 6.]

[22] L. Posani, A. Paccoia, and M. Moschettini, "The carbon footprint of distributed cloud storage," 2019. [Online]. Available: https://arxiv.org/abs/1803.06973 [Page 6.]

[23] E. Blevis, "Sustainable interaction design: Invention & disposal, renewal & reuse," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '07)*. New York, NY, USA: ACM, 2007. doi: 10.1145/1240624.1240705 pp. 503–512. [Page 6.]

[24] C. DiSalvo, P. Sengers, and H. Brynjarsdóttir, "Mapping the landscape of sustainable hci," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. New York, NY, USA: ACM, 2010. doi: 10.1145/1753326.1753625 pp. 1975–1984. [Page 6.]

[25] A. Gustafson, A. Pace, S. Singh, and M. H. Goldberg, "What do people say is the most important reason to protect nature? an analysis of pro-environmental motives across 11 countries," *Journal of Environmental Psychology*, vol. 80, p. 101762, 2022. doi: https://doi.org/10.1016/j.jenvp.2022.101762. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S027249442200007X [Page 7.]

[26] U. Ehsan, P. Tambwekar, L. Chan, B. Harrison, and M. Riedl, "Automated rationale generation: A technique for explainable ai and its effects on human perceptions," 2019. [Online]. Available: https://arxiv.org/abs/1901.03729 [Pages 7, 10, and 59.]

[27] A. Abdul, J. Vermeulen, D. Wang, B. Lim, and M. Kankanhalli, "Trends and trajectories for explainable, accountable and intelligible systems," *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2018. [Page 8.]

[28] T. Miller, "Explanation in artificial intelligence: Insights from the social sciences," *Artificial Intelligence*, vol. 267, pp. 1–38, 2019. [Pages 8 and 59.]

[29] U. Ehsan, Q. V. Liao, M. Muller, and M. O. Riedl, "Expanding explainability: Towards social transparency in ai systems," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021. [Pages 8 and 59.]

[30] M. Mariani, N. Hashemi, and J. Wirtz, "Artificial intelligence-empowered conversational agents: A systematic literature review and research agenda," *Journal of Business Research*, vol. 161, p. 113838, 03 2023. doi: 10.1016/j.jbusres.2023.113838 [Page 8.]

[31] J. Thøgersen, A. Jørgensen, and S. Sandager, "Consumer decision making regarding a "green" everyday product," *Psychology Marketing*, vol. 29, 04 2012. doi: 10.1002/mar.20514 [Pages 8, 10, and 59.]

[32] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: Defining gamification," vol. 11, 09 2011. doi: 10.1145/2181037.2181040 pp. 9–15. [Page 9.]

[33] S. Roeser, "Risk communication, public engagement, and climate change: A role for emotions," *Risk Analysis*, vol. 32, no. 6, pp. 1033–1040, 2012. doi: https://doi.org/10.1111/j.1539-6924.2012.01812.x. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1539-6924.2012.01812.x [Page 9.]

[34] E. D. Mekler, F. Brühlmann, A. N. Tuch, and K. Opwis, "Towards understanding the effects of individual gamification elements on intrinsic motivation and performance," *Computers in Human Behavior*, vol. 71, pp. 525–534, 2017. doi: https://doi.org/10.1016/j.chb.2015.08.048. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0747563215301229 [Page 9.]

[35] E. P. Baumer, "Reflective informatics: Conceptual dimensions for designing technologies of reflection," *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, pp. 585–594, 2015. [Page 9.]

[36] R. Fleck and G. Fitzpatrick, "Reflecting on reflection: Framing a design landscape," in *Proceedings of the 22nd Australian Computer-Human Interaction Conference (OzCHI '10)*. New York, NY, USA: ACM, 2010. doi: 10.1145/1952222.1952269 pp. 216–223. [Page 9.]

[37] E. Ganglbauer, G. Fitzpatrick, and F. Güldenpfennig, "Why and what did we throw out? probing on reflection through the food waste diary," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. New York, NY, USA: ACM, 2015. doi: 10.1145/2702123.2702284 pp. 1105–1114. [Page 9.]

[38] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, pp. 77–101, 01 2006. doi: 10.1191/1478088706qp063oa [Pages 13 and 16.]

[39] B. Fogg, "A behavior model for persuasive design," in *Proceedings of the 4th International Conference on Persuasive Technology*, ser. Persuasive '09. New York, NY, USA: Association for Computing Machinery, 2009. doi: 10.1145/1541948.1541999. ISBN 9781605583761. [Online]. Available: https://doi.org/10.1145/1541948.1541999 [Pages 21 and 59.]

[40] United Nations, "Transforming our world: the 2030 agenda for sustainable development," https://sdgs.un.org/2030agenda, 2015, a/RES/70/1. [Page 21.]

[41] N. Bocken, S. Short, P. Rana, and S. Evans, "A literature and practice review to develop sustainable business model archetypes," *Journal of Cleaner Production*, vol. 65, pp. 42–56, 2014. doi: https://doi.org/10.1016/j.jclepro.2013.11.039. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0959652613008032 [Page 21.]

[42] Y. Huang, A. Skatova, B. Bedwell, T. Rodden, V. Shipp, and E. Bertenshaw, "Designing for human sustainability: The role of self-reflection," in *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct (MobileHCI '15)*. ACM, 2015. doi: 10.1145/2786567.2794323 pp. 1042–1045. [Page 49.]

[43] U. Genç and H. Verma, "Situating empathy in hci/cscw: A scoping review," *Proc. ACM Hum.-Comput. Interact. (CSCW)*, vol. 8, no. CSCW2, pp. Article 513, 37 pages, 2024. doi: 10.1145/3687052 [Page 49.]

[44] N. Zhang, Y. Yan, and R. Suzuki, "From following to understanding: Investigating the role of reflective prompts in ar-guided tasks to promote task understanding," *CoRR*, vol. abs/2501.13258, 2025. doi: 10.48550/ARXIV.2501.13258 To appear in CHI 2025. [Online]. Available: https://doi.org/10.48550/ARXIV.2501.13258 [Page 49.]

[45] C. Satchell and P. Dourish, "Beyond the user: Use and non-use in hci," in *Proceedings of the 21st Australasian Computer-Human Interaction*

Conference (OZCHI). Melbourne, Australia: ACM, 2009. doi: 10.1145/1738826.1738829 pp. 9–16. [Page 50.]

[46] S. Consolvo, D. W. McDonald, and J. A. Landay, "Designing for behavior change in everyday life," in *Computer*, vol. 42, no. 6. IEEE, 2009. doi: 10.1109/MC.2009.199 pp. 86–89. [Pages 50 and 59.]

[47] R. Orji, G. F. Tondello, and L. E. Nacke, "Personalizing persuasive strategies in gameful systems to gamification user types," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI)*. Montreal, QC, Canada: ACM, 2018. doi: 10.1145/3173574.3174009 pp. 1–14. [Page 50.]

[48] E. Luger and A. Sellen, ""like having a really bad PA": The gulf between user expectation and experience of conversational agents," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. San Jose, CA, USA: ACM, 2016. doi: 10.1145/2858036.2858288 pp. 5286–5297. [Pages 51 and 52.]

[49] M. Porcheron, J. E. Fischer, S. Reeves, and S. Sharples, "Voice interfaces in everyday life: How users experience Alexa," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. Montreal, QC, Canada: ACM, 2018. doi: 10.1145/3173574.3174214 pp. 640:1–640:12. [Pages 51 and 52.]

[50] T. W. Bickmore and R. W. Picard, "Establishing and maintaining long-term human-computer relationships," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. Portland, OR, USA: ACM, 2005. doi: 10.1145/1054972.1055030 pp. 454–461. [Pages 51 and 52.]

[51] T. Hu, A. Xu, Z. Liu, Q. You, Y. Guo, V. Sinha, J. Luo, and R. Akkiraju, "Touch your heart: A tone-aware chatbot for customer care on social media," *CoRR*, vol. abs/1803.02952, 2018, presented at CHI 2018. [Online]. Available: http://arxiv.org/abs/1803.02952 [Page 53.]

[52] E. Ortega-Ochoa, M. Arguedas, and T. Daradoumis, "Empathic pedagogical conversational agents: A systematic literature review," *British Journal of Educational Technology*, vol. 55, pp. 886–909, 12 2023. doi: 10.1111/bjet.13413 [Page 53.]

[53] A. Bharadwaj, P. Siangliulue, A. Marcus, and K. Luther, "Critter: Augmenting creative work with dynamic checklists, automated quality assurance, and contextual reviewer feedback," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 2019. doi: 10.1145/3290605.3300769 pp. Paper 539, 12 pages. [Page 54.]

[54] L. D. Setlock, S. R. Fussell, and C. Cruz, "Comparing collaboration and communication among chinese and american distributed teams," in *Proceedings of the 2004 ACM Conference on Computer Supported Cooperative Work (CSCW)*. Chicago, IL, USA: ACM, 2004. doi: 10.1145/1031607.1031705 pp. 568–577. [Pages 55 and 56.]

[55] K. Reinecke and A. Bernstein, "Improving performance, perceived usability, and aesthetics with culturally adaptive user interfaces," *ACM Transactions on Computer-Human Interaction*, vol. 20, no. 2, pp. 6:1–6:29, 2013. doi: 10.1145/2459083.2459087 [Pages 55 and 56.]

[56] C. Nass, Y. Moon, N. Green, B. J. Isbister, and E. Iverson, "Are computers gender-neutral? gender-stereotypic responses to computers with voices," *Journal of Applied Social Psychology*, vol. 27, no. 10, pp. 864–876, 1997. doi: 10.1111/j.1559-1816.1997.tb00275.x [Page 55.]

[57] M. Madsen and S. Gregor, "Measuring human-computer trust," in *Proceedings of the 11th Australasian Conference on Information Systems*. Brisbane, Australia: ACIS, 2000, pp. 6–8. [Page 55.]

[58] K. M. Lee and C. Nass, "Designing social presence of social actors in human computer interaction," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2003. doi: 10.1145/642611.642652 pp. 289–296. [Page 55.]

[59] M. Roshanaei, M. S. El-Nasr *et al.*, "Empathic gaps? evaluating and comparing how GPT-4 and humans express empathy," *arXiv preprint arXiv:2503.00000*, 2025, forthcoming in CHI 2025. [Page 56.]

[60] A. Bandura, *Self-efficacy: The exercise of control.*, ser. Self-efficacy: The exercise of control. New York, NY, US: W H Freeman/Times Books/ Henry Holt & Co, 1997. ISBN 0-7167-2626-2 (Hardcover); 0-7167-2850-8 (Paperback) [Page 59.]

[61] V. Tiefenbeck, T. Staake, K. Roth, and O. Sachs, "For better or for worse? empirical evidence of moral licensing in a behavioral

energy conservation campaign," in *Proceedings of the 21st European Conference on Information Systems*, 2013. [Online]. Available: https://aisel.aisnet.org/ecis2013_cr/102/ [Page 59.]

[62] C. Abraham and S. Michie, "A review of behavior change theories and techniques used in interventions to improve adherence to medication," *Patient Education and Counseling*, vol. 59, no. 1, pp. 17–33, 2005. doi: 10.1016/j.pec.2004.05.025 [Page 59.]

[63] J. Froehlich, L. Findlater, and J. Landay, "The design of eco-feedback technology," *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 1999–2008, 2010. doi: 10.1145/1753326.1753629 [Page 59.]

[64] J. Pierce, W. Odom, and E. Blevis, "Energy aware dwelling: A critical survey of interaction design for eco-visualizations," in *Proceedings of the 28th international conference on Human factors in computing systems (CHI)*. ACM, 2008. doi: 10.1145/1357054.1357322 pp. 2125–2134. [Page 59.]

[65] P. Petkov, F. Köbler, M. Foth, and H. Krcmar, "Motivating domestic energy conservation through comparative, community-based feedback in mobile and social media," in *Proceedings of the 5th International Conference on Communities and Technologies*. ACM, 2011. doi: 10.1145/2103354.2103359 pp. 29–38. [Page 59.]

[66] T. R. Dillahunt, G. Becker, J. Mankoff, and R. E. Kraut, "Ubi-green: Investigating a mobile tool for tracking and supporting green transportation habits," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 2009. doi: 10.1145/1518701.1518861 pp. 1043–1052. [Page 59.]

[67] M. Laschke, M. Hassenzahl, S. Diefenbach, and M. Tippkämper, "With a little help from a friend: A shower calendar to save water," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 2011. doi: 10.1145/1978942.1979036 pp. 633–646. [Page 59.]

[68] J. Hamari, J. Koivisto, and H. Sarsa, "Does gamification work?–a literature review of empirical studies on gamification," *Proceedings of the 47th Hawaii International Conference on System Sciences (HICSS)*, pp. 3025–3034, 2014. doi: 10.1109/HICSS.2014.377 [Page 59.]

[69] B. Knowles, L. Blair, M. Hazas, and S. Walker, "Rethinking plan a for sustainable hci," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*. ACM, 2014. doi: 10.1145/2556288.2557361 pp. 3593–3596. [Page 59.]

[70] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?": Explaining the predictions of any classifier," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016. [Page 59.]

[71] Q. Wang, X. Zhang, and X. Chen, "Interpretability for creative tasks: A case study with ai-generated music," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021. [Page 59.]

[72] M. Hind, Y. Gil, and et al., "Bridging machine learning and design for creative ai tools," in *Creativity and Cognition*, 2021. [Page 59.]

[73] B. Y. Lim, A. K. Dey, and D. Avrahami, "Why and why not explanations improve the intelligibility of context-aware intelligent systems," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2009, pp. 2119–2128. [Page 59.]

[74] D. Z. Wang, A. Abdul, Q. V. Liao, and B. Y. Lim, "Designing human-centered explainable ai: A review and guideline," *Foundations and Trends in Human–Computer Interaction*, 2022. [Page 59.]

[75] B. Knowles, A. K. Clear, M. Van Kleek, R. Binns, D. Fraser, and et al., "Rethinking plan a: The roles and limitations of design in sustainable hci," *Proceedings of the ACM on Human-Computer Interaction*, vol. 2, no. CSCW, pp. 1–21, 2018. [Page 60.]

[76] A. Howes, C. Baber, A. L. Cox, and et al., "Designing ai for emotional support," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020. [Page 60.]

[77] A. Abdul and et al., "Cognitive biases in human-model interaction: A literature review," in *Proceedings of the 2020 ACM Conference on Fairness, Accountability, and Transparency*, 2020. [Page 60.]

[78] M. Bhatt, Q. V. Liao, A. Abdul, and X. Zhang, "Explainable ai for human-centered design: A literature review," *Foundations and Trends in Human–Computer Interaction*, 2021. [Page 60.]

[79] H. Kaur, H. Nori, S. Jenkins, R. Caruana, H. Wallach, and J. Wortman Vaughan, "Interpreting interpretability: Understanding data scientists' use of interpretability tools for machine learning," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020. [Page 60.]

[80] N. Saxena, T. Guha, S. Muralidhar, and et al., "How do people actually use ai explanations? towards explainability-as-a-dialogue," *ACM Transactions on Interactive Intelligent Systems*, 2023. [Page 60.]

[81] D. Shin and Y. Jung, "Beyond pros and cons: Exploring user perceptions of ai explanation interfaces on trust and reliance in ai," *Computers in Human Behavior*, vol. 107, p. 106227, 2020. [Page 60.]



Appendix A

User test plan

User Study Plan

To validate the effectiveness of the improved prototype, a small-scale, exploratory user study will be conducted. Given time constraints and the qualitative nature of this research, the study will focus on **5 participants**, each participating in a test that combines both quantitative and qualitative evaluation components. This current user study adopts a mixed-method approach. (Note: A previous round of fully qualitative testing was conducted separately, based on different tools and prompts, and its results will be analyzed independently.)"

Participant Profile:

The selected participant will be an active creative practitioner with experience using tools such as Midjourney or DALL·E for AI-generated artwork. The participant is recruited based on their familiarity with the topic and their willingness to participate in reflective tasks.

Study Procedure:

Initial briefing (5 minutes): The participant will be informed about the purpose of the study and ethical considerations, including informed consent and data anonymization. The researcher will also introduce the purpose of the prototype, explain how participants are expected to interact with it, and outline the general style of responses the system may generate (e.g., open-ended, sustainability-focused replies). At the beginning of each session, the researcher will share their screen and provide a live, verbal walkthrough of the prototype interface. This includes pointing out where to type input, how to submit messages, and how responses will be displayed. No example prompt will be entered or shown during this introduction, in order to avoid priming effects. The intention is to ensure functional understanding of the interface

while leaving the content of reflection entirely up to the participant.

1. **Interactive session** (15–20 minutes): The participant will interact with the chatbot, which presents a short series of system-generated prompts designed to encourage sustainability reflection. Participants are also encouraged to introduce their own creative context or recent work to make the reflection more personal.

2. **Post-task questionnaire/ interview** (10–15 minutes): This phase includes two components. First, a conceptual quantitative questionnaire is defined to capture core evaluation dimensions for future structured testing. Second, a semi-structured interview was conducted with each participant to explore their subjective experiences with the prototype. These interviews covered areas such as perceived helpfulness, emotional tone, prompt clarity, and integration into their creative process. The qualitative questions were designed to align with the system's three key design specifications and form the basis for the thematic analysis in this study.

Test Instructions

The following instructions were provided to each participant prior to the user study:

Hi, thanks so much for taking the time to help with my research!

This is part of my master's thesis project at KTH. I'm designing a chatbot that helps creative practitioners—like people who use AI tools such as Midjourney or DALL·E—to reflect more deeply on sustainability issues.

Today's session will be short and quite casual. First, I'll briefly explain how the system works. Then you'll interact with the chatbot for around 10–15 minutes. It will ask a few open-ended questions related to sustainability in creative AI work. You can reply in your own words and style. The conversation is open-ended—just see where it takes you.After that, we'll do a quick follow-up interview or questionnaire where I ask for your impressions and feedback.

There are no right or wrong answers—just your honest thoughts and reactions are what matter. Everything is anonymous, and your responses will only be used for academic purposes.

If you're ready, I'll go ahead and show you how it works.

(The user study participant agrees with going ahead)

The chatbot is designed to help you reflect on sustainability in your creative AI work—like when you're using tools such as Midjourney, DALL·E, or ChatGPT.

Questions

Part I: Quantitative Questionnaire (Theoretical Metrics for Future Use)

The following items represent a Likert-style scale used to conceptually map user responses to the system's design strategies. While these were not formally administered during the test, they reflect the evaluative criteria embedded within the conducted interviews and are suggested for future structured evaluation.

Q-A1. The reflection prompts helped me reconsider the sustainability aspects of my work.

Q-A2. The timing of the feedback messages was appropriate and did not feel disruptive.

Q-A3. The emotional tone of the system felt supportive and non-judgmental.

Q-A4. The sustainability feedback was clear, concise, and relevant.

Q-A5. The prompts supported my creative autonomy while inviting reflection.

Q-A6. Overall, the feedback integrated well into the interaction flow.

Scale: [1 – Strongly Disagree] to [5 – Strongly Agree]

(These can be adopted in future iterations for scalable benchmarking.)

Part II: Qualitative Open-Ended Questions (Reflection-Oriented Interview Anchors)

During the user test sessions, the following open-ended questions were either explicitly posed or implicitly explored through guided reflection. They provide qualitative grounding for the evaluation of the prototype in relation to the three core design specifications.

Q-B1. Can you describe any moments where the system encouraged you to reflect on sustainability? What stood out to you?

→ *(Covers Spec 1: Post-generation micro-reflection)*

Q-B2. How did the tone of the system's messages make you feel?

→ *(Covers Spec 2: Empathetic and emotionally supportive prompting)*

Q-B3. What do you think about the clearance and easy understanding of the messages?

→ *(Covers Spec 3: Summary-based, situational sustainability feedback)*

Q-B4. What do you feel about the feel of the workflow combines with the prototype?

→ *(Covers integration into natural workflow, Spec 1 + 3 combined)*

Q-B5. What suggestions do you have for improving how the system communicates sustainability or reflection prompts?

→ *(General improvement feedback – supports all three specs)*

Note: These instruments were not administered as formal questionnaires, but rather informed the structure and analysis of the semi-structured user test conversations. For future evaluations, both quantitative and qualitative variants could be employed to assess system performance more systematically.

Evaluation Focus:

The evaluation will focus on the effectiveness of three core design strategies implemented in the prototype:

1. **Post-Generation Micro-Prompting:** Does the ambient reflection prompt that appears after content generation successfully trigger sustainability reflection without disrupting creative flow?

2. **Empathetic Prompting Tone:** Is the system's tone perceived as emotionally supportive and non-judgmental, and does it enhance user comfort and willingness to engage in reflection?

3. **Summary-Based Sustainability Feedback:** Are the brief, context-sensitive sustainability prompts perceived as timely, helpful, and relevant to the user's current creative task?

These three areas will be explored through a combination of observation, short Likert-scale evaluations, and semi-structured post-session interviews.

Appendix B

Consent form of the User Study

KTH Royal Institute of Technology

Master's Thesis – User Study Consent Form

Title of Study:

Design research case study of LLM-based conversational sustainability reflection tools for Creative AI practitioners

Researcher:

Shaotian Wu (Master's student, Information and Network Engineering, KTH)

Email: shaotian@kth.se

Supervisor:

Petra Jääskeläinen (PhD Candidate, Creative AI Researcher, KTH)

Purpose of the Study

The aim of this study is to evaluate a prototype conversational tool that supports sustainability reflection for Creative AI users, such as those using tools like Midjourney or ChatGPT in their work.

What Participation Involves

You will interact with a conversational prototype tool via a web interface.

The session is expected to last around 10–15 minutes.

After the session, you will be asked a few questions about your experience, either through a short questionnaire or an informal interview.

Voluntary Participation

Your participation is entirely voluntary.

You may stop at any time or withdraw your data without giving a reason.

Data and Anonymity

No sensitive personal data will be collected.

Your responses will be recorded anonymously.

If the session is recorded, it may include screen video or audio, but no facial images or identifying personal content will be collected.

Data will only be used for academic research purposes related to this master's thesis.

No names or contact information will be published.

Contact

If you have any questions about this study, please contact the researcher (Shaotian Wu) or the supervisor (Petra Jääskeläinen).

By signing below, you confirm that you have read the information above and agree to participate in this user study.

Name (printed): ______

Signature: ______

Date: ______

Appendix C

AI Tools Usage Statement

This thesis has employed the assistance of AI tools (such as ChatGPT by OpenAI) during the writing process. These tools were used for language refinement, paraphrasing, and improving the academic tone of certain sections.

An example prompt would be:

”GPT. I am writing the user study part of the thesis. Please help me modify the following paragraph to make it more professional and academic: The user study has shown a positive result for the sustainability reflection chatbot, especially in its use of emotionally supportive and non-judgemental language to increase the sustainability oriented relection. The participants say that the system’s reflective prompts helped them consider about the environmental influence of their work while using a warm and non-intrusive tone. ” The answer output would be:

”Sure! Here’s a revised version of your paragraph with improved academic tone and clarity: The user study revealed a generally positive reception of the sustainability reflection chatbot, particularly in its use of emotionally supportive and non-judgmental language to encourage sustainability-oriented reflection. Participants consistently affirmed that the system’s reflective prompts helped them reconsider the environmental impact of their creative work, while also appreciating the chatbot’s warm and non-intrusive tone. Let me know if you’d like to adjust the level of formality or emphasize specific aspects of the user feedback further.”

These tools are also used for data analysis of the survey results and user study qualitative part analysis. In the analysis of the survey results, these tools are used for analyzing the overall preference of the participants — for instance, comparing average ratings across participants and visualizing responses to questions Q1–Q6 to identify role-based patterns. In the analysis

of the qualitative part of user study, these tools are also used to give an approximate analysis of the ideas from user test participants — for example, summarizing feedback segments by topic, clustering user attitudes toward sustainability prompts, and suggesting thematic labels for preliminary coding.

All content, arguments, and interpretations remain the sole responsibility of the author. The final decisions on wording, structure, and included content were made independently by the author.

TRITA-EECS-EX-2025:699
Stockholm, Sweden 2025

www.kth.se